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# Hybrid Decision Support System for Financial Performance Prediction of SMEs

**Research Project Proposal By**

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بسم الله الرحمن الرحيم

**DEDICATION**

work would not have been possible without Allah grace and guidance. First, I dedicated this work to myself, who persevered, endured, stayed up late, worked hard, and never gave up when surrender was an option. I also dedicate the fruits of my humble efforts to the one who raised me and struggled for me, to the light that illuminated my life, to the one whose name I bear with pride and honor, to the one who encouraged me throughout my life. May you live a long and happy life, my dear father.

I also dedicate this work to my husband and life partner, who supported me every step of the way, patiently and steadfastly. To my child, my family, and all who support me, and to everyone who instilled enthusiasm in me.

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## ABSTRACT

The rapid digital transformation of business environments has generated many amounts of unstructured (textual) data and structure (financial) data. However, Small and medium-sized business (SMEs) financial performance forecasting methods currently in use rely on quantitative financial metrics, without using the value that captured for customer sentiment. This study proposes hybrid decision support system for predicting SME financial performance by combining quantitative financial features with qualitative sentiment features extracted from customer reviews using VADER and FinBERT software. The empirical analysis was conducted on the Olist e-commerce dataset, which includes transaction records and customer reviews for thousands of Brazilian SME vendors.

Three models were built to predict the company's revenue: the frist contain only financial data, the second contains text-data that extracts from customers comment and depends on sentiment analysis, and the third combines the two data sources. Six machine learning algorithms Were applied to all three models: XGBoost, Random forests, Gradient Boosting, Linear Regression, Artificial neural networks and convolutional neural networks. The mean absolute error, root mean square error, and coefficient of determination (R2) were then used to assess the three models' performance.

The results showed that the hybrid model achieved the highest prediction accuracy compared with other models. The Gradient Boosting and Random Frost achieving the highest $R^2$ values of 0.9784 and 0.9720. This confirms that combine textual sentiment features significantly improves the accuracy of SME revenue forecasting. These results contribute to adding knowledge to studies related to AI-assisted financial decision support systems and provide practical applications for SMEs and e-commerce platform operators that search data-driven tools for financial performance forecasting.

## KEYWORDS

Decision Support Systems; Small and Medium Enterprises (SMEs); Artificial Intelligence; Financial Performance; Business Analytics; Financial Data; Textual Data; Sentiment Analysis; NLP; Machine Learning; DL Learning

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## List of Acronyms

| | |
|---|---|
| Small and Medium-sized Enterprises | SMEs |
| Artificial Intelligence | AI |
| Decision Support System | DSS |
| Machine Learning | ML |
| Information Systems | IS |
| Deep Learning | DL |
| Natural Language Processing | NLP |
| Large Language Model | LLM |

# CHAPTER I

# 1.CHAPTER I Introduction

## 1.1 Introduction:

This chapter provides overview of the research topic and presents critical information regarding financial performance analysis in small and medium-sized enterprises (SMEs). Then discusses the research problem, determine the objectives, explains the motivation for conducting for this research, and identifies its scientific contribution. It also shows the research questions and hypotheses that are related to the research topic. Furthermore, it describes the project implementation plan and provides an initial overview of the proposed decision support system.

## 1.2 Overview:

With the rapid development of technology and the increasing reliance on digital business, small and medium-sized enterprises have witnessed rapid growth in recent years [1]. Representing over 90% of global businesses, SMEs are crucial to boosting local economies by creating jobs and contributing significantly to regional GDP [2]. These companies have made remarkable progress in increasing sales rates, leading to economic shifts and fluctuations in the labor market. However, despite their rapid growth, SMEs face difficulties and struggle to meet the challenges of efficient financial resource management and predicting future outcomes. A company's efficiency is measured by its financial performance, according to established accounting standards. This assessment includes indicators such as financial management, operational efficiency, and profit and loss ratios. These factors contribute to improving the efficiency of operational processes. Effective financial planning is important for SMEs to control operating costs and overcome unstable economic [3]. With the increasing complexity of business operations and the resulting massive increase in the volume of data generated daily, companies need financial forecasting to estimate future financial results based on historical data such as revenues, profits, costs, and other financial indicators, as well as anticipated market conditions [4]. Financial forecasting in small and medium-sized enterprises has an effects on the company level and also on national and global economic resilience. It is also one of the most important tools that institutions and investors rely on to estimate future financial performance and make strategic decisions based on accurate information. This has helped decision-makers make accurate financial decisions, engage in strategic planning, and assess risks to achieve precise results [5]. SMEs can also leverage this data to optimize resource allocation, minimize financial risks, and plan for future business expansion [6]. These decisions significantly impact business growth and sustainability [7].

The process of analyzing this data plays a important role in transforming data into useful information that helps in understanding various phenomena and making more accurate decisions. Small and medium-sized enterprises face challenges in data analysis. Generally, data analysis can be defined as the process that contain examining, verifying, selecting, transforming, storing, and modeling data to draw useful conclusions or insights related to future events and to help improve decision-making. The data analysis process follows a specific methodology consisting of four stages: data collection, data processing, data analysis, and understanding the model outputs to evaluate their accuracy and impact on financial decisions [8]. In particular, business analysis is understood as a comprehensive methodology used to identify an organization's needs. It provides solutions to improve organizational performance by optimizing work processes, increasing efficiency, and achieving defined strategic goals. It also extracts insights that help companies understand their performance and develop appropriate strategies. This includes statistical, descriptive, predictive, and prescriptive analysis, each serving specific needs in the decision-making process [9].

Traditional financial data analysis depends on basic statistics to evaluate financial performance. Past data analysis methods involved cleaning spreadsheets, performing queries, and generating reports using tools such as Excel or SQL. However, traditional forecasting tools often fail to meet the needs of small and medium-sized enterprises due to their heavy reliance on manual input and poor data quality that used [10]. However, more advanced forecasting models have surfaced with the development of artificial intelligence (AI) that enables businesses to predict business models and navigate dynamic and volatile business environments [11].

Artificial intelligence is defined as the simulation of human intelligence processes by computer systems that can learn, reason, and solve problems. [12]. The combination of AI and business analytics has led to a radical change in thinking. AI has transformed data analysis from a step-by-step, manual process into a more dynamic one that considers multiple variables, such as seasonal trends and customer behavior. It has also facilitated financial planning for organizations, allowing for the development of detailed scenarios, the detection of anomalies, and the real-time adjustment of financial strategies. With the evolution of business methods, and with the spread of data across multiple media, traditional digital data, such as revenues, profits, return on assets and liquidity ratios, are no longer sufficient on their own for financial forecasting or for evaluating company performance or for providing overview of the enterprise future or the factors affecting it. Therefore, textual data, also known as "unstructured content," is considered an important source for predicting a company's future performance, such as text written in reports, social media, or on their websites [13].

Artificial intelligence (AI) can process both structured and unstructured financial data. AI-powered data analytics relies on a range of techniques to process information and extract insights for faster decision-making. Natural language processing (NLP), which helps understand human language, and the analysis and interpretation of large textual datasets, classifying them as positive, negative, or neutral by using sentiment analysis. It can be used for review social media content or customer comments. Research has shown that sentiment analysis can significantly impact financial performance, providing key insights that enhance and support the understanding of financial data. Additionally, by using This data can predict a company's future performance before it appears in traditional financial results. Machine learning (ML) has various techniques, including support vector machines (SVMs), random forests (RFs), gradient boosting (GB), and decision trees (DT). These methods are ability to capture complex, nonlinear relationships that exist in financial datasets. These models are useful for various applications, including time series prediction, budgeting, expense tracking, resource allocation, bankruptcy prediction and other tasks [14]. These techniques improve the accuracy of financial outcome and reduce bias [15].

Decision Support System (DSS) is the component responsible for financial decision-making. These systems are designed to support decision-making activities in companies or organizations by collecting, processing, and analyzing data [16]. They help decision-makers choose the most suitable option based on the company's available human and financial resources by identifying weaknesses in cash flows and providing accurate forecasts, thus contributing to improved decision quality and reduced uncertainty in the business environment.

Finally, many companies use AI-powered systems to gain a rapid response advantage, mitigate problems, and identify financial opportunities to avoid obstacles. This ensures improved performance and accurate results. For these decisions to be successful, employees must have expertise in analyzing financial data and preparing reliable reports [17]. Excellent human resources possess the ability to analyze financial data, enabling them to make sound decisions that enhance the organization's profitability [18]. In light of the above, it is clear that data analytics and artificial intelligence play a significant role in improving financial forecasting and supporting decision-making processes in organizations. The effectiveness of decision support systems used in companies increases when they analyze different types of data to provide a more overview of company performance and market trends. However, there is still a lack of studies that combine the analysis of different types of data that influence financial decision-making in small and medium-sized enterprises. This study aims to address the existing gap by analyzing these types of data and using them to support forecasting and improve the quality of decisions in the modern business environment.

## 1.3 Problem Statement

Small and medium-sized enterprises (SMEs) are important to the global economy. Enterprises face the challenge of accurately predicting their financial performance to support their strategic decisions. One of the most significant challenges companies face in managing data is amount volume of data generated by their daily operations. It contains both quantitative data (numerical), such as invoices and sales, and qualitative data, which accumulates in the form of non-numerical textual data, such as customer opinions and comments Which are produced through digital sales platforms or social media. This data requires processing to be used for forecasting and decision support to improve a company's operational and market performance compared to other companies, identify potential problems before they occur, and find appropriate solutions.

Despite the existence of many data analytics systems, most are geared towards large enterprises. Most small businesses depend on traditional financial data analysis that neglect qualitative data (such as customer feedback), which has role in financial forecasting, especially in the era of rapid digital transformation. This neglect of qualitative data can create a huge gap, may lead companies to develop solutions that are neither comprehensive nor efficient. Also, leads to a lack of analytical efficiency and the absence of integrated decision support systems.

This research is solving this problem by developing a hybrid predictive model specifically designed for small businesses. The system can process and analyze qualitative and quantitative data types by using many Techniques like machine learning, natural language processing, and deep learning models to understand results and analyses for predicting financial performance and supporting decision-makers in providing comprehensive insights.

## 1.4 Research Goal and Objectives:

**Research Goal:**

To develop an integrated predictive model combining both traditional and textual financial data to improve the ability of small and medium-sized enterprises (SMEs) to predict their financial performance.

**Research Objectives:**

**O1:** To develop a traditional financial forecasting model based on structured financial data.
**O2:** To develop a sentiment analysis-based model using textual data and natural language processing techniques.

**O3:** To design and implement a hybrid model that integrating financial and textual features by using algorithms in machine learning and deep learning.

**O4:** To evaluate three models and compare performance by applying standardized evaluation criteria such as MAE, RMSE, and $R^2$.

**O5:** To analyze the results to provide recommendations that demonstrate how small and medium-sized enterprises (SMEs) can benefit from the most accurate model that is best suited to their operating environment.

## 1.5 Motivation to Conduct Research

The motivation for this research stems be the important role of small and medium-sized enterprises (SMEs) in increasing the economy. Many companies fail within their first-year fill because low revenue, low profits, or other factors. While many studies have analyzed SME data that Widespread in various scientific journals and research papers, most previous studies have focused on only one type of data. This leads to an incomplete picture of the factors affecting company performance, resulting in lower accuracy in analysis and forecasting.

Furthermore, many companies still use traditional analytical methods, which can be time-consuming and labor-intensive, especially given the rapid spread of digital business environments and the increasing number of electronic transactions, which are difficult to analyze using traditional approaches. The availability of Machine Learning (ML) and Deep

Learning (DL) technologies presents an opportunity to enhance forecasting accuracy that supports decision-making processes and enables companies to better understand their future financial performance.

This research also adapts with Saudi Vision 2030, which focuses on digital transformation, artificial intelligence technologies, and supporting a data-driven economy to increased efficiency across various sectors and achieving sustainability and economic growth.

## 1.6 Action Plan

The research will be conducted according to the following plan: The main tasks for conducting this research are divided into 8 phases: (collecting financial data, collecting textual data, building the financial model, training and testing it, building the textual model, training and testing it, building the hybrid model by collecting features from financial and textual data, evaluating the models, comparative analysis, and writing the report).

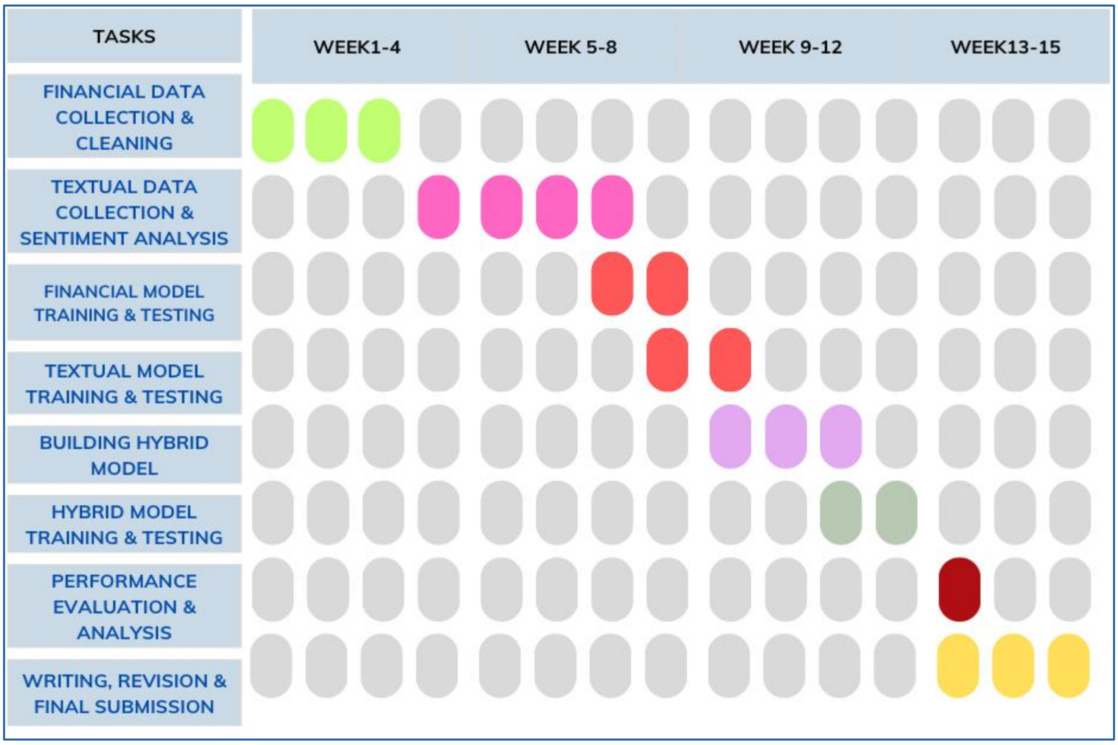



*Figure 1. Gantt chart illustrating the research schedule over 15 week*

Figure illustrates the research project, organized into 8 sequential phases over 15 weeks.

**Stage 1:** Collecting financial data from official sources such as Kaggle and Pre-processing this data.

**Stage 2:** Collecting textual data from official so such as Kaggle and pre-processing this data.

**Stage 3:** Training and evaluating the financial data model using fundamental metrics.

**Stage 4:** Training and evaluating the textual data model according to approved performance standards.

**Stage 5:** Building the hybrid model by combining features of financial and textual data.

**Stage 6:** Training and evaluating the hybrid model using appropriate evaluation criteria.

**Stage 7:** Analyzing and evaluating performance, comparing this study with previous studies in this field, and interpreting the results.

**Stage 8:** writing the final research.

## 1.7 Research question:

The following research questions are the focus of this study:

**Main Question:**

How can build a hybrid predictive model that combines financial data and sentiment analysis that is derived from customer feedback to predict the financial performance for small and medium-sized enterprises (SMEs)?

**Sub-questions:**

- To what extent does a hybrid model, combining financial data and sentiment analysis, outperform financial models in predicting SME revenues?
- Can textual sentiment data alone predict SME revenues without the need for financial data?
- Which predictive model achieves the highest accuracy in forecasting SME revenues?
- Which technique is more effective when used for predictive financial performance: Machine Learning (ML) or Deep Learning (DL)?

## 1.8 Significance and Impact of Research

The importance of this study enhancing the prediction of the financial performance of small and medium-sized enterprises (SMEs) by using both financial indicators and customer sentiment. That improves the competitiveness of companies by monitoring changes in customer opinions and market behavior earlier than traditional method for financial analysis methods, allowing companies to respond more quickly and make appropriate decisions. It also makes it easier for managers and employees to understand the company's future trends.

### 1.8.1 Research Contributions

The study contributes to improves the field of financial forecasting for SMEs. This addresses a gap identified in previous studies by presenting an integrated model for predicting the financial performance of small and medium-sized enterprises (SMEs) through combines financial indicators with customer sentiment extracted from textual data such as customer comments. The integration model helps to understand how qualitative information impacts financial forecasts, helping to enhance the competitive advantage of SMEs and support decision-making. Various artificial intelligence and natural language processing techniques, such as FinBERT, are applied to extract sentiment attributes (positive, negative, neutral) for identifying the most efficient model for predicting financial performance and improves understanding of algorithm performance by comparing several algorithm from machine learning and deep learning models. This study also offers a practical and scientific contribution applicable to similar data on other e-commerce platforms. Furthermore, it can be developed for future research in the fields of management, finance, and data analysis.

### 1.8.2 Potential Value of Research:

The research adds value to (SMEs) for Enhancing the accuracy of forecasting financial performance by:

- Developing solutions that help SMEs make informed decisions based on accurate models, rather than dependent on traditional methods
- Helping companies understand customer opinions and satisfaction with services or products at an early stage.
- Supporting strategic planning for businesses.
- Enhance for SMEs competitiveness and sustainability.

## 1.9 Hypothesis / Null Hypothesis

- **Main Hypothesis**

  $H_{01}$: Integrating textual sentiment data that derived from customer reviews and financial data does not improve accuracy of SME revenue prediction

  $H_{11}$: Integrating textual sentiment data that derived from customer reviews and financial data improves accuracy of SME revenue prediction

- **Sub-Hypothesis 1 (Model Performance):**

  $H_{02}$: The hybrid model does not outperform the financial-only model in prediction performance ($R^2$, MAE, and RMSE).

  $H_2$: The hybrid model outperforms the financial-only model in prediction performance ($R^2$, MAE, and RMSE).

- **Sub-Hypothesis 2 (Sentiment Method Comparison)**

  $H_{03}$: FinBERT-derived sentiment features do not provide higher predictive value than VADER-derived sentiment features.

  $H_3$: FinBERT-derived sentiment features provide higher predictive value than VADER-derived sentiment features.

## 1.10 Block Diagram of Proposed Approach

The Figure below explains the research process that show series of logically ordered steps that illustrate the study's implementation. The diagram consists of six main, color-coded stages for easy identification. The process starts with data collection from reliable sources such as Kaggle (Step 1). This is followed by feature identification (Step 2). In Step 3, three models are built: a financial model, a textual model, and a hybrid model. In Step 4, evaluated all models by using appropriate metrics. Finally, Step 6 performs a comparative analysis by comparing proposed models.

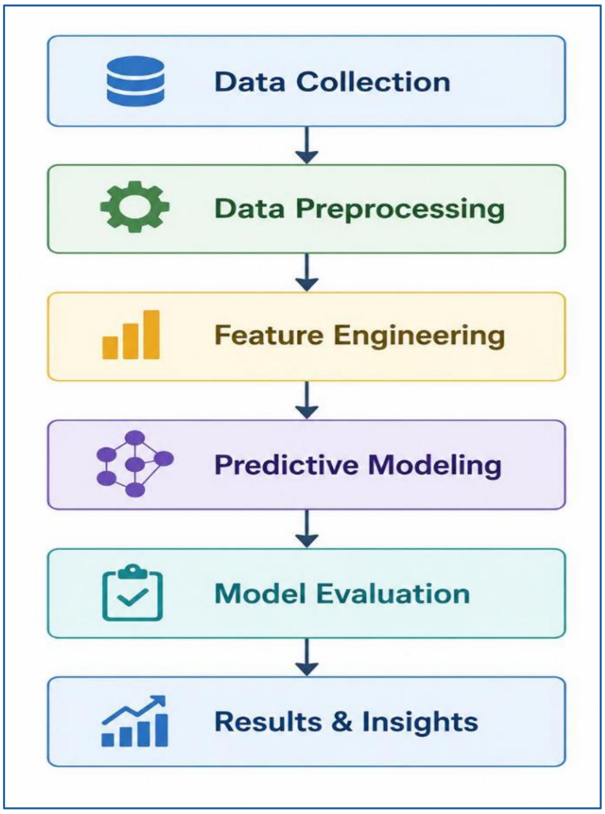


*Figure 2 Research process*

## 1.11 Thesis Structure

The structure of this study is as follows:

**Chapter 1:** Introduction – outlines overview of the project, defines the research problem, clarifies its objectives, and highlights the contribution of the study.

**Chapter 2:** Literature Review – provides background information and reviews previous studies related to the analysis of financial and textual data.

**Chapter 3:** Theoretical framework – Discussion foundation of the study. It also reviews the concept of artificial intelligence and machine learning, the rationale for selecting the algorithms, and explains the proposed framework.

**Chapter 4:** Research Methodology – explain details the dataset used (financial and textual data), the preprocessing operations, the feature extraction procedures, how the three models (financial, textual, and hybrid) were constructed, and the mechanisms for their evaluation.

**Chapter 5:** Results – This chapter show descriptive statistics and results of the algorithms that used to building models and conducts a comparative analysis of the models used.

**Chapter 6:** Discussion – This chapter presents an explanation of each model used, a comparison between machine learning and deep learning models, and the results in light of existing literature.

**Chapter 7:** Conclusion and Future Studies – This section summarizes the main findings of this thesis, the limitations of the study, and offers a set of recommendations for future study.

# CHAPTER 2

# 2. CHAPTER II Literature Review

## 2.1. Introduction

This chapter overviews the research concepts, including the small and medium-sized enterprises (SMEs), financial performance and Financial Data Analysis Techniques, textual data analysis in the financial field, and sentiment analysis. In addition, the chapter covers a selection of previous studies related to the research topic and discussing Barriers, Challenges, and Limitations in Existing Research. Also, explain the relationship of the research in the Saudi Arabian Context concludes by discussing the research gaps.

## 2.2 Small and Medium Enterprises (SMEs):

Small and medium-sized enterprises (SMEs) are defined as businesses with less than 250 workers and limited revenue. SMEs are important for increase economic that contribute to create job and supporting economic development both locally and globally. Recent business and economics literature indicates that this type of company constitutes the largest proportion of establishments in most economies, representing approximately 90% of all businesses worldwide.

SMEs possess several characteristics that prefer them to large companies, most notably their high degree of flexibility in responding to market changes and their ability to innovate and adapt to the times. However, approximately twenty percent of small enterprises can fail in their first year , and only 50% still after five years may lack funding or low revenue, making financial performance management and forecast crucial for their continued success. Also, Previous literature indicates limited access to advanced data analysis systems that support decision-making. In addition, their dependence on traditional financial data methods makes limits their ability to analyze and predict future performance compared to larger companies with more sophisticated analytical capabilities.

## 2.3 Financial Performance of SMEs

Financial performance is defined as process aimed at evaluating the past, present, and projected financial position of organizations and suggesting potential areas for improvement. Measurable indicators are used to evaluate the effectiveness and financial utility of enterprise. These indicators are extremely useful in planning and development to ensure business growth and strength. Financial performance evaluation depends on several indicators that provide a clear picture of operational efficiency and profitability. This achieved by using tools such as financial ratio analysis, including profitability ratios (gross profit and net profit), liquidity ratios, debt ratios (such as the debt-to-assets ratio), efficiency ratios, return on investment ratios, and others, which measure a company's efficiency and financial performance.

Studies indicate that forecasting financial performance is crucial for decision-making in small and medium-sized enterprises (SMEs) that enables decision-makers to understand how to invest and allocate financial resources, identification risks, and identify the company's strengths and weaknesses. Furthermore, it helps improve future planning [20]. This contributes to informed decision-making and allows managers to make strategic adjustments to improve the financial position, thereby boosting company growth and increasing its competitiveness in the market.

### 2.3.1 Financial Data Analysis Techniques:

processed and analyzed financial data to extract knowledge and transform it into useful information for prediction through three main stages: preprocessing, data extraction and data visualization.

Initially, data is preprocessed by cleaning it, correcting missing values, standardizing its formats, and reducing noise before being fed into analysis models to improve data quality. The study previous indicated that preprocessing improves the accuracy of analysis, thus contributing to enhanced reliability of the final results [21].

Then, data extraction stage involves identifying patterns and relationships within the financial data that importance of these methods in financial forecasting [22]. Then, using data mining techniques that depend on artificial intelligence and machine learning technologies. These techniques include classification, regression, clustering, and association rules. Classification helps in assessing financial risks such as credit risk. Regression is used to predict future financial values such as revenues and profits. Clustering helps in dividing customers into different categories based on their financial behavior, while association rules are used to discover relationships between different financial indicators [23].

Finally, using data visualization, which presents the analytical results through graphs, dashboards or interactive reports. This process clarifying financial trends and revealing anomalies within the data that helps decision-makers better understand the data [24]

## 2.4 Textual Data Analysis in the Financial Domain

Textual data is unstructured information that contain customer opinions, comments, reviews. In recent years, this type of data has gained attention due to digital development and increasing reliance on it. Analysis of textual data depends on many techniques that transform text into analyzable data by using To extract patterns and meaning from the text, use Natural Language Processing (NLP). This is a crucial step to reduce noise data by removing unnecessary symbols and punctuation, lowercase text, stop words, and tokenization. This way improves data quality before building models. Then transformed text into numerical data (feature extraction).

one of the most crucial techniques used to analyze text is TF-IDF (Term Frequency - Inverse Document Frequency) measures the importance of a word within the text. The textual data analysis has several challenges, including the lack of data structure, the diversity of languages and dialects, the presence of spelling errors, and the difficulty in understanding context

## 2.5 Financial Sentiment Analysis

Sentiment analysis is a part of textual research that used to extract subjective information, specifically opinions, by determining the emotional tone underlying a text to measure the emotions expressed in the words [25] by uses statistics, natural language processing, and machine learning to extract information from text and known as emotional artificial intelligence [26]. In the context of small and medium-sized enterprises (SMEs), sentiment analysis is considered as important tool and a key component of business intelligence [27]. It also plays a role for enhancing customer engagement [28], which results from Comments, interactions, and ratings on social media are essential for businesses to improve their operations and achieve sustainable growth, as well as to make decisions that align with customer needs

Furthermore, when a problem arises, sentiment analysis helps clarify it before it escalates [29]. It also helps in making decisions that align with customer requirements.

This analysis also helps in discovering trends in social networks and reflecting reality [30]. Emotions in content are typically categorized as positive, negative, or neutral. Various methods can be used to analyze sentiment, including lexical analysis, that relies on classifying words in the text and the number of frequency, using a dictionary containing words categorized as positive or negative according to their intensity [31]. Another type of analysis is ontological analysis, where build models that classify emotions based on concepts [32].

With advancements in artificial intelligence technologies, deep learning-based models have emerged that offering high accuracy in text analysis.

One of these techniques is FinBERT, which built a modes based on the BERT architecture and specifically trained on financial data. This model understanding sentence context by analyzing the full meaning of the text. Then categorizes emotions as positive, negative, or neutral, from analyzing customer feedback and textual data related to financial performance.

## 2.6 Related Work (Previous Studies)

### 2.6.1 Financial-Based Studies

Zainol Abidin et al. (2020) [33] The effectiveness of logistic regression models and artificial neural network (MLP) models was measured for predicting the failure of small and medium-sized enterprises (SMEs). The study was conducted on 41 failing and 41 successful companies to identify and highlight the influential indicators that lead to a company's success or failure. The results showed the superiority of artificial neural networks over logistic regression in classifying successful and failing companies. ANN achieved a predictive accuracy of 98.2%, compared to 86% for logistic regression. However, these results and studies were subject to several limitations. The study was conducted on a relatively small sample and within a short timeframe, as the data collected in the past two years was insufficient to detect failure, develop strategic recovery plans, and take the necessary measures to improve the company's profitability and liquidity. Therefore, a longer observation period, such as five years prior to failure, should be adopted to obtain more practical and highly accurate results.

Patalay and MadhusudhanRao (2019) [34] This study proposed developing an intelligent system based on machine learning techniques to analyze financial data and extract patterns that aid in predicting stock prices. This system considers the fundamental characteristics of companies and macroeconomic factors to improve prediction accuracy. The neural network model was trained using open-source financial data for a number of stocks listed on the Bombay

Stock Exchange (BSE) and the National Stock Exchange of India (NSE). The results showed that using artificial neural networks to analyze financial data helps predict stock prices up to one month in advance. Furthermore, the study indicated that the proposed decision support system assists decision-makers and enables individual investors to make more accurate decisions at a lower cost compared to hiring financial advisors. Although the system relies on integrating the intrinsic characteristics of stocks with external macroeconomic factors, thus improving prediction accuracy, the study's scope is limited to specific stock data from the BSE and NSE, which restricts the generalizability of the results. Further development of the machine learning models is also needed to keep pace with dynamic market changes.

Cuervo (2023) [35] developed a model for predicting the financial performance of small and large companies using artificial intelligence (automated regression models and neural networks) and Bayesian models to forecast the financial performance of both small and large businesses. To improve prediction accuracy, a new set of financial and macroeconomic ratios were added to traditional financial ratios. 46,000 financial data observations were used to train and test the models. The results showed that neural networks (FNNs) excel at extracting complex relationships between financial indicators, particularly when predicting more complex ratios such as operating cash flows and profit margins. Bayesian networks, however, can outperform in specific conditions. The study confirmed that feature engineering and careful variable selection enhance model accuracy. Bayesian networks, however, could outperform other models under specific conditions. From the above, the Bayesian approach requires extensive expertise in network structure and probability distributions, which limits the possibility of rapid generalization.

Tong and Tian (2023) [36] This study proposed an intelligent financial decision support system using ETL technologies, Big Data and Web Crawler. This system can extract and analyze external and internal company data to build a data warehouse sharing platform that support collective business for decision-making based on the relevant data. It also connects this platform to a cloud computing platform for financial report analysis. The system allowed for real-time tracking of the business's financial performance and made use of important financial metrics like profit, return on net assets, and accounts receivable. The results showed that system improved the presentation of financial indicators and increased the number of monthly financial reports. This resulted in a reduction in the time required to analyze it. The study confirmed that integrating big data into financial decision support systems significantly improved corporate financial management, enhanced the effectiveness of financial decisions, and made practical contributions to corporate financial decision-making. The study based on a single case study, which restricts the generalizability of the findings, and lacks validation across multiple datasets or diverse financial environments.

Zhang (2025) [37] created a big data based financial data analysis and decision support system to improve the precision and effectiveness of financial decision management in challenging market conditions. The system was implemented on a single company, and key indicators like cost of sales, profitability, current assets, sales revenue and liabilities, total company assets, and shareholders' equity were selected. Hadoop and Spark big data tools were used to process the large datasets quickly and efficiently, providing accurate early warnings of financial risks and scientific support for informed investment decisions. The study results show that the system operates with high efficiency in terms of data processing speed, accuracy, and user satisfaction. Accuracy reached 98.9%, and user satisfaction increased from 8.2 to 8.7 over three years, improving the effectiveness of financial management and the precision of organizational decision-making processes. Despite the importance of the study, its application was limited to a single company and relied on internal factors without considering external and market factors. Furthermore, the system was not tested over a longer period to ensure its sustainability. The system's functionality needs to be further enhanced, and its applications expanded to adapt to the evolving financial management requirements of organizations.

Elkenawy (2026) [38] proposes new technique to enhance the accuracy of the financial model for making financial decisions in small and medium enterprises, such as revenues, profit margins, and cash flows, by applying the Logarithmic Transformation (LogTrans) model as a basis for forecasting, and then improving it through several modern Metahorist algorithms and selecting the most influential features, including the Simulated Annealing (SSO) algorithm, the Particle Swarm Optimization (PSO) algorithm, the Whale Optimization (WAO) algorithm, and others. The optimized SSO + LogTrans model outperformed the other models, achieving a remarkable mean squared error of $1.95 \times 10^{-7}$, a root mean squared error of $4.42 \times 10^{-4}$, and a high $R^2$ value of 0.966. The study also indicated that using metahorizontal optimization algorithms significantly improved the prediction accuracy and generalizability financial decision-making models for small and medium-sized enterprises (SMEs). While the study improved model accuracy, it focused solely on numerical financial data and did not integrate other data sources such as textual data or customer opinions. Therefore, there is still a need to develop hybrid predictive models that combine these elements.

Gupta (2025) [39] create prediction model for sales by using deep learning techniques . The study processed the data and extracted key characteristics such as order count, average purchase value, and delivery time. A neural network model was then applied to improve revenue forecasting accuracy. The results shown the of outperforme deep learning models in capturing many patterns within the data. However, the study focuse on structured data and did not integrate textual data, such as customer comments into the prediction models or using natural language processing or sentiment analysis techniques. This limits the models ability to understand qualitative aspects of customer behavior.

### 2.6.2 Sentiment and Textual Analysis Studies

He (2025) [41] developed a hybrid model for analyzing customer opinions in small and medium-sized enterprises (SMEs). This model integrates textual data with images from customer reviews to improve accuracy compared to traditional text-only methods. The Hybrid Multimodal Model was used, including BERT for semantic meaning extraction from the texts, BiLSTM (Bidirectional LSTM) for word sequence analysis, Attention Mechanism for assigning greater weight to the most important words or features, and Visual Feature Extraction for extracting features from the images. The results proved the hybrid model achieved higher accuracy than traditional text-based sentiment analysis models. Despite the positive results, the study presents several challenges, including the complexity of integrating different data types (text and images) and the need for more efficient models for handling large datasets.

Li (2025) [42] explored the potential of using sentiment analysis in marketing for small and medium-sized enterprises (SMEs) and how such analysis can help companies improve their marketing strategies and enhance customer communication to make data-driven decisions. The study employed an analytical approach by analyzing customer comments on social media to understand their opinions about services and products. A dictionary-based method, relying on a dictionary of positive and negative words, was used Machine learning algorithms such as

Naïve Bayes and Support Vector Machine (SVM) were employed. The study results showed that machine learning provided higher accuracy in classifying sentiments compared to dictionary-based methods. The study also revealed that most current sentiment analysis tools are unsuitable for local languages or informal texts used on social media.

Orji et al. (2022) [43] analyzed customer opinions posted on Twitter to improve Business Intelligence (BI) systems in Nigerian SMEs. The study aimed to extract information that would help companies understand customer satisfaction and opinions, leading to better business decisions. It involved collecting a large number of tweets related to companies or products and then processing them by removing irrelevant symbols and words and converting the text into an analyzable format. Sentiment analysis technique was applied by using Natural Language Processing (NLP) to categorize the tweets positive, negative, or neutral. The results showed that sentiment analysis from Twitter provides valuable information about customer opinions, and companies. This data can use to improve their decision-making processes. This helps enhance marketing strategies and customer service. The study also demonstrated that integrating sentiment analysis with Business Intelligence systems helps SMEs better understand the market. However, the study highlighted some research limitations. It focused primarily on textual data from Twitter without incorporating other data sources such as financial data or actual sales figures. Furthermore, it did not utilize advanced Machine learning models to forecast future business performance.

Umarani et al. (2021) [44] Analyzed and compared sentiment analysis methods using machine learning and deep learning to determine the efficiency of each model and the possibility of selecting the most suitable technique for classifying texts in different fields. A restaurant review dataset containing 1000 reviews was used, and several machine learning algorithms were applied: Naive Bayes, logistic regression, Supporting Vector Machine (SVM), Random Forest, Nearest Neighbor (KNN), and Decision Tree. Additionally, deep learning was applied using convolutional neural networks (CNNs) and long short-term memory (LSTM). The models were evaluated using classification measures, such as accuracy, F1-score, AUC, ROC Curve, Precision and Recall. According to the findings, the Naive Bayes model outperformed the other Machine Learning models in terms of AUC (~0.76) and was the quickest to train. Deep Learning achieved higher accuracy, with CNN reaching approximately 84% and LSTM reaching approximately 82%, but they took longer to train. The study concluded that Deep learning models excel in accuracy, while machine learning models excel in speed and efficiency. However, the study was limited to a small textual dataset and did not incorporate other data sources such as financial data.

Karanikola et al. (2023) [45] This study systematically compared Machine Learning algorithms with several pre-trained Deep Learning models for financial sentiment analysis, applied to a widely used handwritten dataset in financial sentiment research. Several techniques were applied, including SVM, Naive Bayes, and Random Forest, with deep learning models such as BERT, RoberTa, and three variants of FinBERT. FinBERT is a financial-focused language model built on the BERT architecture, trained to analyze financial texts and extract sentiment from market and investment news. It undergoes two phases: pre-training on massive financial datasets (billions of words) and fine-tuning on specialized financial sentiment analysis tasks.The results revealed the clear superiority of FinBERT models over general BERT and all classical approaches, particularly in complex financial contexts requiring a deep understanding of specialized terminology, thanks to its pre-training on large and specialized financial datasets. Therefore, the study was limited to data from large companies listed on financial markets, and these models were not tested on texts produced from the SME environment, such as customer comments, social media platforms, and online reviews. This constitutes a clear gap in assessing the suitability and efficiency of these models when applied to informal textual data from small and medium-sized enterprises.

### 2.6.3 Summary of Previous Studies

| Author(s) | Years | Objective | Dataset used | Techniques | Key Findings | Research Gap |
|---|---|---|---|---|---|---|
| **1. Financial-Based Studies** | | | | | | |
| Patalay and | 2019 | Design an ANN-based financial | Historical stock market data | Linear Regression, ANN with | ANN significantly outperformed Linear Regression due to its | The study was confined to organised stock |

| Author(s) | Years | Objective | Dataset used | Techniques | Key Findings | Research Gap |
|---|---|---|---|---|---|---|
| Madhusud hanRao | | decision support system for stock price prediction in volatile markets | from BSE/NSE (India) | multiple hidden layers | pacity to model non-linear relationships in financial time series | markets with abundant historical data, making it inapplicable to SMEs that lack such structure |
| Cuervo | 2023 | Predict financial performance of SMEs and large enterprises using newly proposed financial and macroeconomic ratios | Financial + macroeconomic data (mixed sources) | ML Regressors, Neural Networks, Bayesian Models | ANN outperformed all other models across six prediction tasks including ROA, ROE, and net margin | There is a lack of standardized performance evaluation metrics and no consistent reporting of numerical accuracy, which limits comparability across different predictive models. |
| Z. Abidin et al. | 2020 | Compare the capability of Logistic Regression and ANN in predicting SME failure | dataset of 82 SMEs (41 failed, 41 non-failed) from the hospitality | Logistic Regression, ANN | ANN achieved a predictive accuracy of 98.2% compared to 86% for Logistic Regression, confirming the superiority of deep | The sample size is very small, which limits the generalizability of the results. |

| Author(s) | Years | Objective | Dataset used | Techniques | Key Findings | Research Gap |
|---|---|---|---|---|---|---|
| | | in the Malaysian hospitality sector | sector in Malaysia | | learning over traditional statistical models | |
| Zhang | 2025 | proposed an intelligent financial decision support system incorporating risk early warning and investment decision | One Company financial data | Big Data Analytics, Web Crawler, ETL Pipeline Quantitative | The system demonstrated improvements in user satisfaction and processing speed for financial risk identification and investment decision support | The study based on a single case study, which restricts the generalizability of the findings, and lacks validation across multiple datasets or diverse financial environments. |
| L. He | 2025 | Examine the influence of accounting system quality and human resource competence on financial | Survey data (150 SMEs, North Medan) | Survey, Multiple Regression Analysis | Both accounting system quality and HR competence significantly and positively affect the quality of financial decisions in SMEs | The study is based on survey data only, without real operational or transactional data, and lacks predictive modeling |

| Author(s) | Years | Objective | Dataset used | Techniques | Key Findings | Research Gap |
| --- | --- | --- | --- | --- | --- | --- |
| | | decision-making quality in SMEs in North Medan | | | | components, limiting the robustness of the findings. |
| D. Tong and G. Tian | 2023 | Build an intelligent financial decision support system that automates financial analysis by combining big data with the Internet Plus platform. | J Group financial + industry web-crawled data | Big Data, Web Crawler, ETL, Internet Plus Platform | The system improved automation and efficiency in financial decisicision-making, reducing dependence on human judgement | The system focuses mainly on data processing efficiency rather than predictive intelligence, making it more descriptive than a true intelligent decision-support system. |
| Elkenawy | 2026 | Enhancing SME financial | SME financial dataset | LogTrans + Metaheuristic | The optimized model greatly increases forecasting | Limited focus on model optimization |

| Author(s) | Years | Objective | Dataset used | Techniques | Key Findings | Research Gap |
|---|---|---|---|---|---|---|
| | | forecasting accuracy using metaheuristic-optimized ML models | (unspecified source) | Optimization (PSO, WOA, SSO) | accuracy $R^2 = 0.966$ compared to baseline models. | without developing new predictive frameworks |
| **2 - Sentiment and Textual Analysis Studies** | | | | | | |
| Orji et al. | 2022 | Improve SME business intelligence using sentiment analysis | Twitter dataset (SME-related tweets – Nigeria) | BNB, SVC, Logistic Regression | Logistic Regression achieved 83% accuracy | The study's dependence on Twitter, a single social media platform, limits generalizability and creates statistical bias. |
| A. Karanikola et al. | 2023 | Compare classical ML vs transformer models in financial sentiment analysis | Financial text data (news/social media – not specified) | Classical ML vs BERT, RoBERTa, FinBERT | Transformer models outperform classical ML | Lack of integration with financial forecasting systems |
| Li..Z | 2025 | Improve customer engagement | Customer feedback + behavioral | Sentiment analysis + | Improved engagement, | Limited real-world validation and |

| Author(s) | Years | Objective | Dataset used | Techniques | Key Findings | Research Gap |
|---|---|---|---|---|---|---|
| | | using sentiment + predictive modeling | data (not specified) | predictive ML models | prediction, and customer loyalty | scalability issues |
| L. He, | 2025 | SME e-commerce reviews dataset (text + images – custom collected) | Improve sentiment analysis using multimodal data fusion (text + image) | BERT-BiLSTM-Attention + CNN (multimodal fusion) | Multimodal model achieves higher accuracy than unimodal approaches | Limited generalization due to custom dataset and lack of large-scale validation |
| Umarani et al. | 2021 | Compare ML vs DL techniques for sentiment classification | Standard sentiment dataset (not specified) | NB, SVM, RF, KNN, CNN, LSTM | Deep learning models outperform ML models | Lack of real-world domain application and deployment |

*Table 1 summary of literature*

## 2.7 Barriers, Challenges, and Limitations in Existing Research

Despite improvements in business research on financial performance predictions and the Artificial intelligence application, including machine learning and deep learning approaches, most previous studies face several challenges.

1. The limited availability of real-world data for small and medium-sized enterprises (SMEs). This data is important for developing predictive models, unlike large enterprises that publish annual reports and profit disclosures. This leads to search for available alternative data sources. As Tong and Tian [34] demonstrated, the absence of data reduces the performance of machine learning models in accurate prediction.

2. Another challenge is quality of available data. Many studies have problems such as missing values, data imbalances and noise. This may affect forecasting performance. orji et al. [36] indicates these challenges directly impact the accuracy of predictive models.
3. Using different evaluation metrics and methodologies across studies make comparisons of results difficult. Also, differences in application environments may affect the generalizability of models.
4. The difficulty of generalization, meaning that prediction results vary depending on the type and size of data. Some studies indicate the Machine Learning models such as GB, Random Forest and XGBoost Outperforms over traditional models, while other studies show the opposite. Deep Learning models achieve high accuracy, but they need a lot of data to achieve reliably.

## 2.8 The Subject in the Saudi Arabian Context

Saudi Arabia is concerns on Artificial Intelligence to transform infrastructure development that aligning with Saudi Vision 2030. This transformation supports both public and private sector, one of these sectors is financial sector. Predicting the financial performance for small and medium-sized enterprises (SMEs) is important element in increasing economy and reducing reliance on oil. Despite this progress, research specifically focused on predicting the financial performance of SMEs in Saudi Arabia using AI remains limited. Most Previous studies have been writing in other countries, such as Brazil, China, and Western nations, with little research specifically addressing the Saudi business environment. Furthermore, there is no found research that integrating textual data, such as customer reviews, with financial data into predictive models within the Saudi context.

With the rapid growth of e-commerce platforms in Saudi that customers reliance on them for purchases, resulting in financial data such as profits and revenues, as well as textual data such as customer comments and ratings, the Saudi context presents a significant opportunity for developing research in the field of AI-powered financial forecasting. While this study utilized the Brazilian Olist dataset as the model application due to data availability, the proposed hybrid framework can be extended to include similar Saudi platforms such as Noon, Salla, and Zid, which offer comparable data environments. Future research can validate and validate these models using Saudi-specific datasets.

## 2.9 Research gap

Recent studies in the field of financial performance forecasting for Small and Medium-sized Enterprises (SMEs) have shown clear evolution in forecasting models. These models began with traditional statistical models and have since developed into more sophisticated models capable of detecting complex and nonlinear financial patterns. This depend on Artificial Intelligence like Machine Learning, Deep Learning and Natural Language techniques. However, these methodological developments have revealed three interconnected research gaps that limit the effectiveness and applicability of current models.

The first gap is the complete reliance on traditional financial data as the basis for predictive modeling.

For example study [46] relied on traditional financial models using statistical methods such as linear regression, ARIMA models, and exponential normalization. the study demonstrated that traditional methods are insufficient for understanding the financial complexities of SMEs. Similarly, study [34] showed that linear regression is inadequate for predicting risks based on traditional attributes such as company size, age, international expansion, and risk management [9]. Some studies have focused on adding external factors to enhance the predictive capabilities of traditional models. For example, a study by Malakauskas and Laxotin demonstrated that the random forest model achieved an accuracy of 85% to 92%, outperforming logistic regression.

This is achieved by enhancing the effectiveness of financial models through the addition of factors such as time, credit history, and company age. Similarly, a study by Cuervo [35] shows that the accuracy of predictive models is enhanced by integrating new financial variables such as ROA, ROE, net margin, and operating margin with traditional indicators. Most studies show the superiority of machine learning and deep learning models over traditional statistical methods. However, all these studies rely entirely on the assumption that financial performance can be explained solely by quantitative data, which may not provide a clear picture of the modern business environment. They overlook external factors such as customer behavior and opinions, which can significantly influence financial results and shape future performance. While parallel research has addressed the importance of text analysis, studies such as those by tong and tain [36], orji et al. [42], and li [41] have demonstrated that extracting sentiment from financial texts provides substantial and independent predictive value. The not integrate these insights into financial forecasting models highlights a clear gap between two evolving research paths that remain largely unconnected.

The second gap to the lack of studies designed especially for small and medium-sized enterprises (SMEs). Most studies rely on data from large, publicly traded companies. Much of the literature developing sentiment analysis and hybrid modeling techniques depends on formal textual sources such as earnings calls, annual reports, and analyst disclosure resources that are largely unavailable to SMEs. For example, study tong and tian [36] focus on large companies restricted the generalizability of their results by analyzed sentiment using the HowNet dictionary with a deep learning model, but their depend on a Chinese-language dictionary limited the generalizability of their findings. In contrast, other studies focused on SMEs such as Zainol Abidin et al [33], and Zamil [40] , but they offered a systematic review or questionnaire without applied model. This highlights a clear lack of practical predictive models.

The third gap is lack of a hybrid predictive framework that integrates financial data and textual analysis within the context of SMEs. Although some studies have focused on financial modeling, and others have addressed textual data separately, attempts to integrate these two data sources into a single model remain very limited. This gap is becoming increasingly important in e-commerce environments where small and medium-sized enterprises (SMEs) generate large amounts of textual data through customer reviews and interactions, which remains underutilized in financial forecasting models.

# CHAPTER 3

# 3. CHAPTER 3 Theoretical Framework

## 3.1 Introduction:

In this chapter shows theoretical and conceptual framework of the research. It explains the intellectual framework justifying the chosen methodology, which integrates quantitative financial data with qualitative textual data to forecast the revenues of small and medium-sized enterprises (SMEs). The first section focuses on theoretical framework, relying on three theories found in the literature. These theories show the importance of textual data and role in providing informational value that financial data alone cannot offer. The second part shows the foundations of Artificial Intelligence and Machine learning, which will be used to build the predictive model. The final section focuses on explaining the reasons for choosing the algorithms used in this research, how evaluated these models, and presenting the proposed conceptual model. This section illustrates how the research problem can be transformed into an applied model that can be implemented and tested.

## 3.2 Theoretical Framework

This research based on three theories that explain the relationship between textual data and the financial data in SMEs, that highlight the importance of integrating both types of data into a unified predictive framework.

### 3.2.1 Digital Economy Theory

With rapid digital development, the business environment has been radically reshaped. With rapid digital development, the business environment has been radically reshaped. Traditional financial data is not sufficient for prediction alone. Also, Unstructured data, such as customer ratings, comments, and reviews play important role in determining customer satisfaction with services and products. Digital economy theory demonstrates the importance of digital data in forecasting and decision-making, especially in this context. Tapscott [46] demonstrated that the modern economy relies heavily on digital data, while Brynjolfsson and McAfee [47] showed that companies using both structured and unstructured data achieve better performance compared to companies that rely solely on traditional data. In the context, customer reviews can be considered an important source of digital data, as they are continuously generated and reflect real-life customer experiences and opinions.

### 3.2.2 Signaling Theory

Signaling theory is important in explaining the significance of textual data and its ability to add independent predictive value to financial performance. This theory was developed by Spence [48]by understanding how textual data can reflect important information that benefits businesses, this theory posits that parties with more information can send signals to other parties to clarify their true situation. Spence [48] also clarified that signals are not limited to formal reports but can include any behavior or information that reflects reality.

In the e-commerce environment, customer comments can be considered signals that reflect the quality of products and services like Positive comments indicate customer satisfaction with the company's performance, while negative comments can affect financial performance.

### 3.2.3 Behavioral Finance Theory

Kahneman and Tversky [49] introduced behavioral finance theory, which explains that economic decisions are not only depends on financial indicators but are also influenced by individual behavior and psychological factors. The theory demonstrates that individuals do not make entirely rational decisions, but their decisions may be also influenced by emotions and impressions. Barberis and Thaler [50] demonstrated that the collective emotions of consumers significantly impact purchasing decisions and the financial performance of companies. Based on These three theories complement each other in providing a clear explanation of the importance of using textual data alongside financial data that provides a more comprehensive and accurate view for interpreting and predicting financial performance. Therefore, this research suggests that a hybrid model that integrating financial data with sentiment analysis derived from textual data can more accurate results compared to using each data type separately.

## 3.3 Artificial Intelligence:

(AI) is a technology that makes it possible for machines to carry out a variety of tasks in a manner like how humans think. In the corporate context, AI helps improve performance, increase competitiveness and plays a significant role in increasing operational efficiency, expanding capabilities through text recognition, customer service, and behavior analysis. It also enables the development of predictions based on this data. Furthermore, AI helps companies make accurate decisions using data generated from their daily operations.

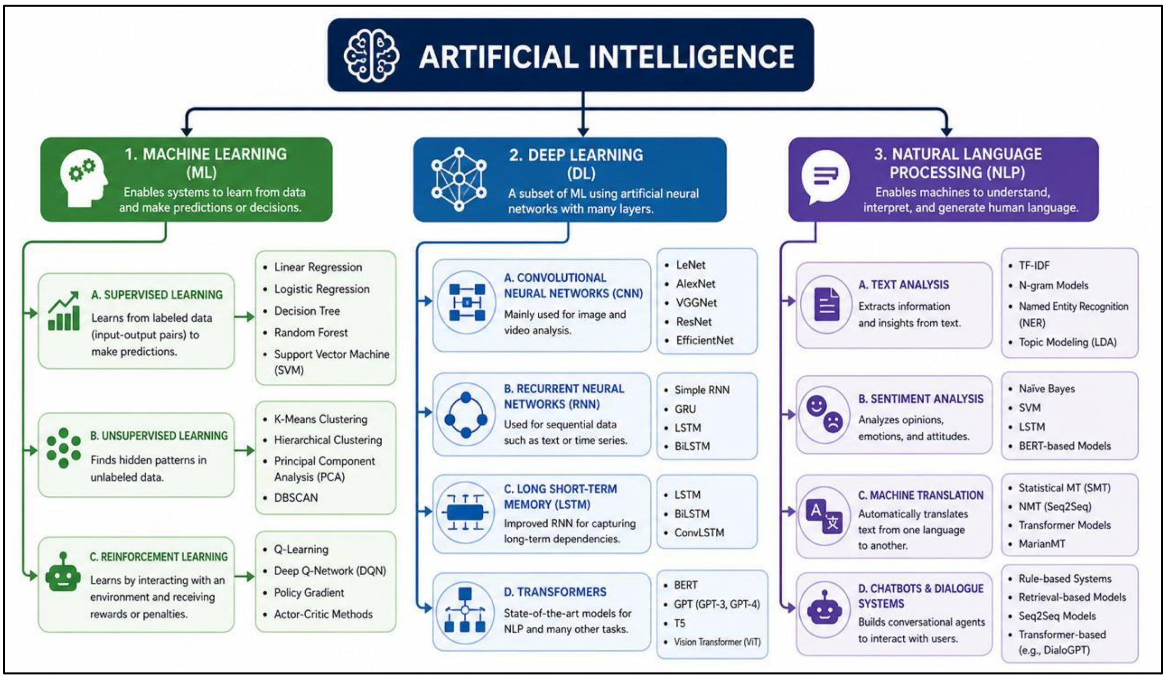



*Figure 3 AI concept*

### 3.3.1 Machine learning:

The method of programming machine learning is classified

**- Supervised learning**

is one of the simplest types of machine learning. In this method, algorithms are trained on a dataset, giving the algorithm a basic understanding of the problem. The data points to be processed are identified, and the algorithm can then easily determine the relationship between inputs and outputs. It is used for classification and regression (prediction) [51].

**- Unsupervised Learning:**

Learning occurs without a training set. Algorithms explore patterns among data points in an abstract way, without human intervention. It is used for clustering and correlation [52]

**- Reinforcement Learning:**

The algorithm learns from new situations using experience and rewards.

In this research, supervised learning was employed by training models using data containing predefined inputs (features) and outputs (target variables). Specifically, regression was used to predict continuous numerical values such as revenues by learning the relationship between a set of independent variables, which include financial characteristics and sentiment indicators, and the dependent variable, which represents financial performance.

### 3.3.2 Deep Learning:

It is developed type from machine learning that depends on artificial neural networks and derived from biological neural networks like brain. These networks are distinct by their ability to learn complex data representations by taking data, training themselves to recognize patterns, and then predicting the output of a new set of similar data.

Deep learning models are used to predict revenue by analyzing the complex relationships between financial and textual variables. study [24] demonstrated that deep learning makes significant useful in extracting patterns from data, especially in environments with large datasets. In this context, Artificial Neural Networks (ANNs) are used in financial modeling and their ability to learn complex relationships between data. It has proven their ability to predict stock returns with higher accuracy compared to linear models. In addition, deep learning models such as CNN and LSTM have replaced traditional statistical methods and machine learning algorithms for forecasting tasks, thanks to their exceptional ability to predict financial time series data and ability to handle time data and extract long-term patterns.

## 3.4 Algorithm selection

In this research, six algorithms (linear regression, random forest, Gradient Boosting, XGBoost, artificial neural networks, and convolutional neural networks) were selected to build a comprehensive experimental framework that allows for the evaluation of the performance of a variety of predictive models.

### First: Basic Statistical Models

Linear regression was used as a baseline model to compare the performance of other models, understand the nature of the data, and determine if there was a linear relationship between the independent and dependent variables. Linear regression is characterized by its ease of implementation, rapid training, and the ease with which results can be interpreted.

**Second: Machine Learning Models**

Random forest, Gradient boosting, and XGBoost were chosen for their high capacity to handle nonlinear relationships and detect complex interactions between variables.

These models are also characterized by their ability to reduce prediction errors and improve accuracy, especially in business data.

Zain Al-Abedin [33] demonstrated that machine learning models, such as the random forest model, outperformed traditional regression models in predicting the financial performance of small and medium-sized enterprises (SMEs).

The Gradient Boosting model was selected based on a systematic review by Zamel [40], which demonstrated its effectiveness in financial forecasting.

The XGBoost model was chosen because Zamel [40] proved effective in addressing the tabular data challenges faced by small and medium-sized enterprises (SMEs), such as data limitations and imbalances.

**Third, deep learning models:**

such as artificial neural networks and convolutional networks, were used due to their ability to extract complex patterns from data, especially when combining different data types, such as financial and textual data, a capability further developed by Tong & Tian [36]. Batalai & MadhusudanRao [34] demonstrated the superiority of artificial neural networks over linear regression models in financial forecasting. Their ability to generate nonlinear outputs across multiple layers makes them suitable for solving complex financial regression problems involving correlated financial and textual variables.

Therefore, this multi-algorithm presentation aims to compare traditional and advanced models and identify the most efficient model for forecasting monthly revenue.

### 3.5 Conceptual Framework:

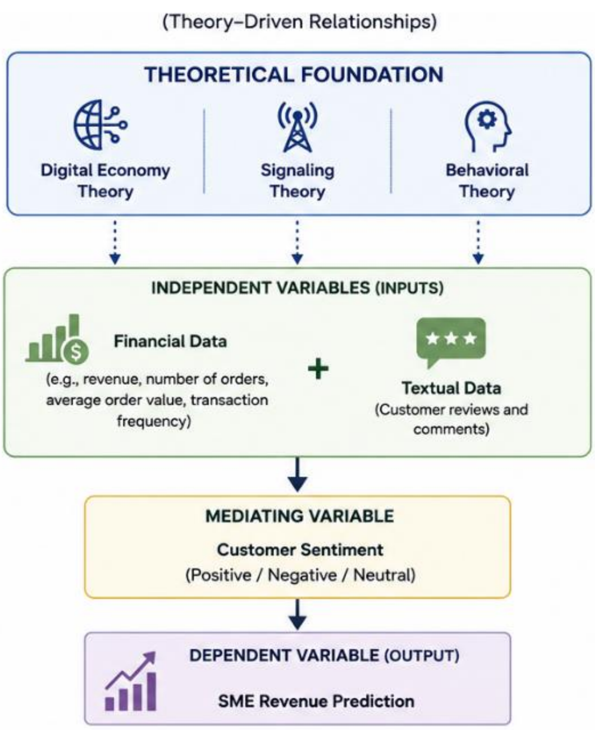



*Figure 4 conceptual framework*

The conceptual framework for this study based on three main theories that shown above: digital economics theory, signal theory, and behavioral theory. These allow to understand how financial or textual data affects revenue prediction for small and medium-sized enterprises (SMEs).

From digital economics theory, financial data such as revenue, order volume, and transaction frequency, serves as a key indicator of company performance in digital markets. This reflects and explains the operational efficiency and economic growth of SMEs, directly impacting revenue projections.

From Signal theory can explain the importance of textual data derived from customer reviews and comments. That can Converting information about product quality and service performance to reduces information lack of symmetry between sellers and customers, indirectly affecting company revenue.

From Behavioral theory, can support this relationship by exploring how customer perceptions, emotions, and satisfaction levels affect from their behavior by using sentiment analysis that extract from textual data, where customer emotions (positive, negative, or neutral) act as a mediating variable between textual comments and financial performance.

Accordingly, the conceptual framework suggests that financial data has a direct impact on revenue forecasts for small and medium-sized enterprises (SMEs), while textual data affects revenue indirectly through customer sentiment. This mechanism show the importance of integrating quantitative and qualitative data sources to better understand and predict business performance in the digital economy.

# CHAPTER 4

# CHAPTER 4 Methodology

## 4.1 Introduction:

This chapter explains the data collection and processing methods and how to build the three predictive models. Also, explain the aims of research methodology used in this study, which focuses on analyzing and predicting revenue

The chapter is divided as follows: Section 1 reviews the research design, methodology, and study context. Section 2 describes the dataset used and its characteristics. Section 3 explains how to process both types of data. Section 4 explains how to extract the independent financial and textual variables. Section 5 reviews the construction and training of the three models. Section 6 explains the framework used, and the chapter concludes with the adopted evaluation strategy. The importance of this chapter lies in providing a clear and reproducible methodology for subsequent researchers to apply the same framework to similar data and verify the results or generalize them to different contexts.

## 4.2 Research design

The design of the proposed research framework consists of sequential stages that aim to improve accuracy of forecasting the financial performance for small and medium-sized enterprises (SMEs) by integrating financial and textual data. This framework is based on three main predictive models: the financial model, the textual model, and the hybrid model, as illustrated in Figure .

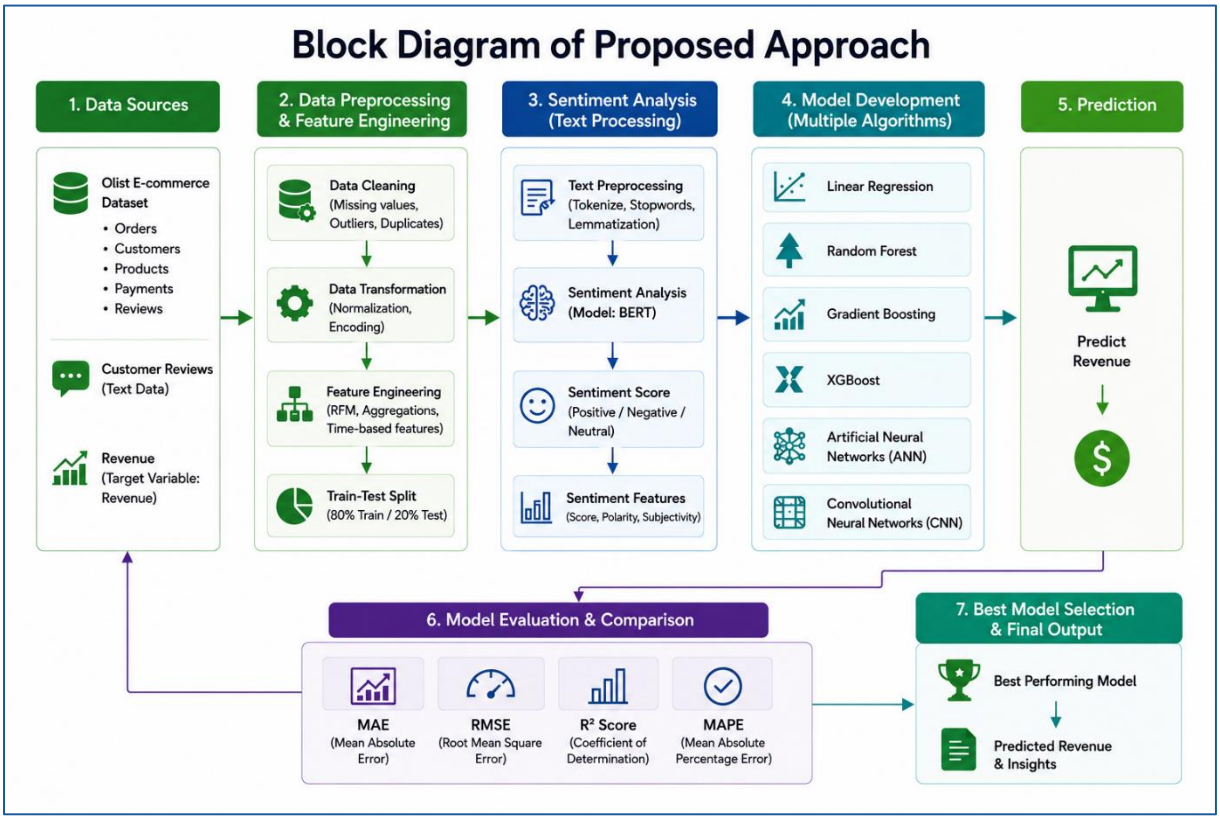



*Figure 5 Research design*

**Stage 1**

This stage involves data integration, where two different types of data are extracted.

The first type is structured financial data, that includes indicators such as revenue, number of orders, average order value, and transaction frequency, that reflect the company's actual performance from a financial perspective.

The second type is unstructured textual data that is extracted from customer reviews and comments, that reflects user experience and satisfaction levels with products and services. This data is analyzed by using sentiment analysis techniques, such as advanced language models, to transform it into quantitative indicators. These indicators include the degree of sentiment (positive or negative).

**Stage 2**

This stage involves feature representation by transforming the dataset into a set of features. The financial features include indicators such as total revenue, order frequency, average order value, and sentiment indicators.

**Stage 3**

This stage involves building and training three different models:

First, the Financial Model, which depend on financial data. This model serves as a baseline for measuring performance, reflecting traditional financial forecasting methods.

Second, the Textual Model, which depend on sentiment indicators that extracted from textual data to assess the ability of data alone to explain financial performance.

Third, the Hybrid Model, which represents the core contribution of this research, combines financial and sentiment data into a single model to improve forecast accuracy.

These models are trained using many of prediction algorithms, including machine learning (ML) and deep learning (DL) models.

**Stage 4**

this stage involves performance evaluation, where each model is evaluated by using set of regression measures, including MAE, RMSE, and $R^2$, allowing for comparison of all algorithms.

**Stage 5**

This stage involves comparative analysis by compared performance of the hybrid model with base model based on financial data. This aims to measure the added value that textual data provides in improving prediction accuracy.

### 4.2.1 Methodology

This methodology aims to define the general framework of the practical procedures followed in the study, which will help organize the work and address various problems to achieve the desired objectives.

**Research Methodology and Reasons for its Selection:**

The methodology used in this research is a quantitative experimental approach based on comparing multiple predictive models.

Based on our study, which aims to develop a hybrid predictive model combining financial and qualitative models for revenue forecasting, it is appropriate to adopt the quantitative experimental approach.

This approach depend on measurable numerical data, enabling the application of statistical and machine learning techniques in analysis and pre. Although the dataset includes customer evaluation data in textual form, these evaluations are converted into quantitative variables such as rating scores or sentiment-based numerical representations. Therefore, the textual data is not analyzed qualitatively but rather transformed into structured numerical characteristics suitable for the built models.

The data was selected from a readily available dataset to support the study. This dataset contains financial information as well as textual information such as customer comments, making it easily accessible and ensuring the clarity of the content required for analysis.

### 4.2.2 Study Context

This research targeted small and large companies, specifically choosing the e-commerce sector as an applied field in this study because it is considered a rich environment of digital data that reflects company performance through commercial transactions. E-commerce data becomes a scientific reflection of company activities such as sales, orders, payments, and customer reviews, which facilitates the analysis of their financial performance and the forecasting of revenues. It is also a suitable model for applying machine learning techniques and predictive models.Therefore, the use of e-commerce data has become a suitable choice for the operational and financial behavior of companies in a modern digital environment.

## 4.3 Dataset Preparation

### 4.3.1. Dataset Description

The dataset used in this study is the Brazilian E-Commerce Public Dataset by Olist. It contains 99441 rows collected over a period of two years. The dataset captures various aspects of online purchasing behavior, including transaction details, payment information, delivery timelines, and customer feedback. All identifying information has been anonymized.

The data is organized across multiple relational tables. The orders table contains the primary structure, including order identifiers, customer identifiers, timestamps, and order status. The order items table provides product-level details such as item prices and product identifiers. The payments table includes the monetary value associated with each transaction, while the reviews table contains both numerical ratings and textual feedback provided by customers.

This multi-table structure enables a comprehensive view of each transaction but requires careful integration to construct a unified analytical dataset.

The following tables were extracted from the dataset:

- **orders_dataset:** Contains order identifiers, customer identifiers, order status, and all relevant timestamps (purchase, approval, carrier delivery, customer delivery, estimated delivery).
- **order_items_dataset:** Links orders to products and contains item-level prices.
- **order_payments_dataset:** Provides order identifiers and the payment_value field, which served as the target variable (revenue).
- **order_reviews_dataset:** Contains order identifiers, customer review scores (review_score), and the review text (review_comment_message).
- **Products_dataset:** contains the information details about the products.

### 4.3.2. Target Variable Definition

The target variable for the prediction task is revenue. Revenue is defined as the total payment amount associated with each order. Since an order may include multiple payment entries due to installments or multiple transactions, the payment values were aggregated at the order level. This aggregation ensures that each order corresponds to a single revenue value, maintaining consistency across the dataset. The resulting target variable reflects the actual monetary value generated from each transaction and serves as the dependent variable for all predictive models.

### 4.3.3 Data Integration

Data integration was performed by selecting the orders dataset as the primary reference table. All additional data sources were merged into this base using left join operations. This approach ensured that all original orders were preserved, even when corresponding data in other tables were missing.

Maintaining the full dataset was a critical decision. In many real-world datasets, not every order includes reviews or complete product details. Using restrictive joints would have significantly reduced the number of observations and introduced bias into the modeling process. By preserving all orders and handling missing values separately, the dataset retained its original scale and diversity.

### 4.3.4. Data Preprocessing

Data preprocessing was carried out to ensure consistency, reliability, and suitability for modeling. The process included handling missing values, transforming data types, and preparing variables for analysis.

**Handling Missing Values**

Missing values were treated differently depending on the nature of each variable.

- For the target variable, observations with missing revenue values were removed. Since the target represents the outcome to be predicted, imputing it would distort the learning process and lead to unreliable models.
- For numerical features, missing values were replaced using the median of each variable. This choice was made because financial data often contains extreme values, and the median provides a stable representation of the central tendency without being influenced by outliers.
- Certain variables, such as the number of products sold, represent counts. In such cases, missing values were interpreted as zero, reflecting the absence of recorded items rather than random missingness.

- Categorical variables were standardized by assigning a consistent label for missing entries. This allowed the models to treat missing categories as a distinct group rather than ignoring them.
- Textual data required a different approach; missing review texts were replaced with empty strings, and additional features were created to indicate whether a review was present. This ensured that the absence of text was explicitly captured as information rather than discarded.

**Temporal Data Processing**

Time-related features were transformed into structured variables suitable for analysis. Timestamp fields were converted into date-time formats to allow arithmetic operations.

- Delivery time was calculated as the difference between the purchase date and the delivery date. This variable reflects operational efficiency and plays a significant role in customer satisfaction.
- Additional features such as the day of the week and the month were extracted from the purchase timestamp. These features capture temporal patterns and seasonal effects that influence purchasing behavior.
- Missing values in delivery time were replaced with the median value to maintain consistency across observations.

**Handling Duplicates and Data Inconsistencies**

In addition to handling missing values, a dedicated step was performed to identify and resolve duplicate and inconsistent records within the dataset.

Duplicate entries were examined at different levels of granularity. At the order level, the order_id was expected to be unique in the final analytical dataset. Any duplicated records resulting from joins across multiple tables were carefully handled through aggregation operations rather than direct removal. For example, payment records associated with the same order were aggregated to compute a single revenue value per order. Similarly, product-level records were grouped to derive features such as the number of products sold and average price.

In cases where exact duplicate rows appeared due to data merging or data collection issues, these records were removed to prevent bias in model training. This ensured that no observation was overrepresented in the dataset.

Data inconsistencies were also addressed. These included:

- Invalid or illogical date sequences, such as delivery dates occurring before purchase dates. Such records were either corrected when possible or excluded from calculations involving time-based features.
- Negative or zero values in financial fields, which were inspected and handled appropriately depending on their context.
- Outliers in numerical features were not removed directly but were mitigated through the use of robust statistical measures such as median imputation and model selection techniques less sensitive to extreme values.

Additionally, categorical variables were standardized to ensure consistency in labeling. Variations in category naming were unified to avoid duplication of semantic categories.

These steps ensured that the dataset used for modeling was clean, consistent, and representative of real-world scenarios, reducing the risk of biased or misleading results.

The final Dataset after the preprocessing is shown in the figures

| | order_purchase_timestamp | order_delivered_customer_date | payment_value | num_products_sold | avg_price | total_price | review_score | delivery_time | day_of_week | mo |
|---|---|---|---|---|---|---|---|---|---|---|
| count | 17713 | 17209 | 17713.000000 | 17713.000000 | 17713.000000 | 17713.000000 | 17713.000000 | 17713.000000 | 17713.000000 | 17713.00 |
| mean | 2017-12-31 08:04:29.057189632 | 2018-01-14 10:56:29.628043264 | 88.488926 | 1.029244 | 88.687629 | 97.839381 | 4.798651 | 12.016542 | 2.784904 | 6.03 |
| min | 2016-09-04 21:15:19 | 2016-10-11 14:46:49 | 0.000000 | 1.000000 | 3.850000 | 3.850000 | 1.000000 | 0.000000 | 0.000000 | 1.00 |
| 25% | 2017-09-12 13:29:37 | 2017-09-25 21:42:41 | 0.000000 | 1.000000 | 79.000000 | 86.800000 | 5.000000 | 6.000000 | 1.000000 | 3.00 |
| 50% | 2018-01-19 12:09:56 | 2018-02-03 03:17:23 | 38.220000 | 1.000000 | 79.000000 | 86.800000 | 5.000000 | 10.000000 | 3.000000 | 6.00 |
| 75% | 2018-05-04 16:17:14 | 2018-05-15 23:54:49 | 115.030000 | 1.000000 | 79.000000 | 86.800000 | 5.000000 | 15.000000 | 4.000000 | 8.00 |
| max | 2018-09-20 13:54:16 | 2018-09-21 23:46:29 | 13664.080000 | 12.000000 | 3105.000000 | 13440.000000 | 5.000000 | 191.000000 | 6.000000 | 12.00 |

*Figure 6 Data after processing*

```
data .info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99441 entries, 0 to 99440
Data columns (total 16 columns):
 #   Column                         Non-Null Count  Dtype
---  ------                         --------------  -----
 0   order_id                       99441 non-null  object
 1   customer_id                    99441 non-null  object
 2   order_status                   99441 non-null  object
 3   order_purchase_timestamp       99441 non-null  datetime64[ns]
 4   order_approved_at              99281 non-null  object
 5   order_delivered_carrier_date   97658 non-null  object
 6   order_delivered_customer_date  96476 non-null  datetime64[ns]
 7   order_estimated_delivery_date  99441 non-null  object
 8   payment_value                  99441 non-null  float64
 9   num_products_sold              99441 non-null  float64
 10  avg_price                      99441 non-null  float64
 11  total_price                    99441 non-null  float64
 12  review_score                   99441 non-null  float64
 13  delivery_time                  99441 non-null  float64
 14  day_of_week                    99441 non-null  int32
 15  month                          99441 non-null  int32
dtypes: datetime64[ns](2), float64(6), int32(2), object(6)
memory usage: 11.4+ MB
```

*Figure 7 Data information*

### 4.4 Feature Engineering

Feature engineering was conducted to transform raw data into meaningful inputs for the models. Three distinct feature sets were developed corresponding to the three modeling approaches.

#### 4.4.1. Financial Model Features

The Financial Model relied exclusively on transaction, product, and temporal features aggregated at the order level.

- num_products_sold: Count of distinct products per order.
- avg_price: Mean price across all items in an order.
- delivery_time: Calculated days between purchase and customer delivery.
- day_of_week and month: Derived from purchase timestamp.
- order_status: Categorical values encoded numerically using scikit-learn's LabelEncoder.

#### 4.4.2. NLP Model Features (Two Versions)

**Text Cleaning:** For all NLP components, review text was cleaned by converting to lowercase and removing all non-alphabetic characters using the regular expression library.

**NLP Model Version 1 using VADER:**

- sentiment_score: Compound sentiment score generated by the VADER (Valence Aware Dictionary and sEntiment Reasoner) algorithm.
- sentiment_label: Binary classification where scores ≥ 0 were labelled positive (1) and others negative (0).
- review_length: Word count of the cleaned review text.
- is_positive: Binary indicator derived from the sentiment score.
- TF-IDF vectors: Generated using TfidfVectorizer

**NLP Model Version 2 using FinBERT:**

- finbert_score: Continuous sentiment score calculated as the probability difference between the positive and negative classes from the FinBERT model (ProsusAI/finbert), a transformer pre-trained on financial sentiment.
- is_positive: Binary indicator derived from the FinBERT score.
- review_length: Word count of the cleaned review.
- sentiment_review_interaction: An interaction term calculated as finbert_score * review_score.

**4.4.3. Hybrid Model Features**

The Hybrid Model combined selected financial and NLP features into a single feature matrix:

- **Financial features:** num_products_sold, avg_price, total_price, review_score,

deliery_time, day_of_week, month.

- **NLP features (VADER):** sentiment_score, review_length, is_positive.
- **Review indicator:** has_review, a binary feature indicating whether a customer provided a written review.

### 4.5. Model Building and Assessment

#### 4.5.1. Data Splitting and Scaling

For all experiments, the dataset was split into a training set (80%) and a testing set (20%) using a fixed random state of 42 to ensure reproducibility. Numerical features were standardized using Standard Scaler to achieve zero mean and unit variance. For the financial and hybrid models, a Simple Imputer with median strategy was applied prior to scaling.

#### 4.5.2. Machine Learning Models

The following supervised learning algorithms were implemented using scikit-learn and XGBoost libraries:

- **Linear Regression:** Served as the baseline model.
- **Random Forest Regressor:** Ensemble of decision trees.
- **Gradient Boosting Regressor:** Sequential ensemble.
- **XGBoost Regressor:** Optimized gradient boosting implementation.

#### 4.5.3. Deep Learning Models

Two neural network architectures were built using TensorFlow/Keras:

- **Artificial Neural Network (ANN):** A fully connected network with ReLU activation on hidden layers and a single linear output node.
- **Convolutional Neural Network (CNN):** A 1D CNN with a single convolutional layer (32 or 64 filters, kernel size 2), followed by a flattening layer and dense layers. Input data was reshaped to (samples, features, 1) to accommodate the Conv1D layer.

#### 4.5.4. Model Training Configuration

- Machine learning models were trained using default parameters, with additional configurations (n_estimators=300, max_depth=15) applied to the Random Forest and XGBoost models in the hybrid setup.

- ANN and CNN models were compiled with the Adam optimizer and mean squared error (MSE) loss function. Training ran for 20 to 30 epochs with a batch size of 32 or 128. No early stopping was used to maintain focus on comparative performance.

### 4.6. Proposed Framework

This study was conducted in a phased approach to evaluate the contribution of different feature sets to revenue prediction accuracy.

**Stage 1: Financial Model Development**:

The Financial Model was developed using only transaction, product, and temporal features. This stage established a performance baseline using traditional structured data.

**Stage 2: NLP Model Development (Two Sub-stages)**

- **Sub-stage 1 (VADER):** Text-based features were extracted from customer reviews using the VADER sentiment analysis tool. TF-IDF vectors were also generated and combined with sentiment features to train a separate set of models.
- **Sub-stage 2 (FinBERT):** The VADER sentiment analysis was replaced with the FinBERT transformer model. A new feature set, including the (finbert_score) feature and an interaction term with (review_score) feature , was generated. TF-IDF features were omitted in this stage.

**Stage 3: Hybrid Model Development**

The Hybrid Model was constructed by concatenating the financial feature set with the NLP feature set from the VADER-based extraction. The (has_review ) feature indicator was added to explicitly capture the presence of textual feedback.

### 3.6. Evaluation Metrics

All models were evaluated on the 20% testing hold-out set using three standard regression metrics:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual revenue.
- **Root Mean Squared Error (RMSE):** Penalizes larger prediction errors more heavily.

- **R-squared (R2):** Indicates the proportion of variance in the target variable explained by the model.

# CHAPTER 5

# CHAPTER 5 Results

## 5.1 Introduction:

This chapter presents the experimental results obtained from the analysis and implementation of three categories of models: Financial Models, NLP-based Models (two versions), and Hybrid Models. The objective of this chapter is to provide a structured comparison of model performance using standard regression metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared ($R^2$).The results are organized into tables for clarity, followed by detailed interpretation and comparative analysis of each model category.

## 5.2 Exploratory Data Analysis (EDA)

This part presents the exploratory data analysis conducted on the Olist Brazilian E-Commerce Public Dataset. The analysis aims to uncover underlying patterns, distributions, relationships, and anomalies within the data before proceeding to predictive modeling. Each visualization is accompanied by a descriptive summary highlighting key findings and their business implications.

### 5.1.1 Data Integration

| Table | Columns |
|---|---|
| orders_dataset | order_id, customer_id, order_status, timestamps |
| order_items_dataset | order_id, product_id, seller_id, price |
| order_payments_dataset | order_id, payment_value, payment_type |
| order_reviews_dataset | order_id, review_score, review_comment_message |
| products_dataset | product_id, product_category_name |

| Table | Columns |
|---|---|
| customers_dataset | customer_id, customer_city, customer_state |

*Table 2 Dataset After merging*

The analysis began by merging six distinct dataset tables: orders, order items, payments, reviews, products, customers. The primary key relationships across these tables enabled the creation of a unified analytical dataset containing transactional, product, customer, and review information at the order level. The merged dataset contained 99,441 unique order identifiers before filtering for complete payment information.

### 5.1.2 Data Visualization

The Figure presents **a** monthly revenue trend line chart that displays the aggregated sum of order prices across each month from September 2016 to October 2018.


*Figure 8 Monthly Sales Trend*

The following chart illustrates the number of distinct orders placed each month throughout the dataset's temporal coverage


*Figure 9 Orders per Month*

The following pie chart displays the proportion of orders completed using each available payment method: credit card, boleto (Brazilian bank voucher), debit card, and voucher.

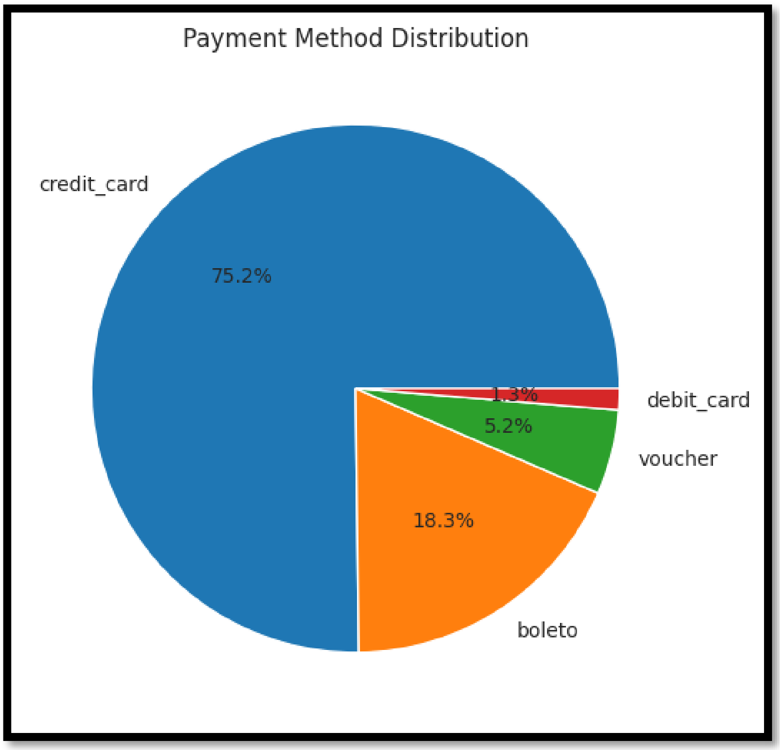



*Figure 10 Payment Type Distribution (Pie Chart)*

The following horizontal bar chart ranks product categories by their frequency of occurrence in the order items dataset, showing the ten most commonly purchased categories.

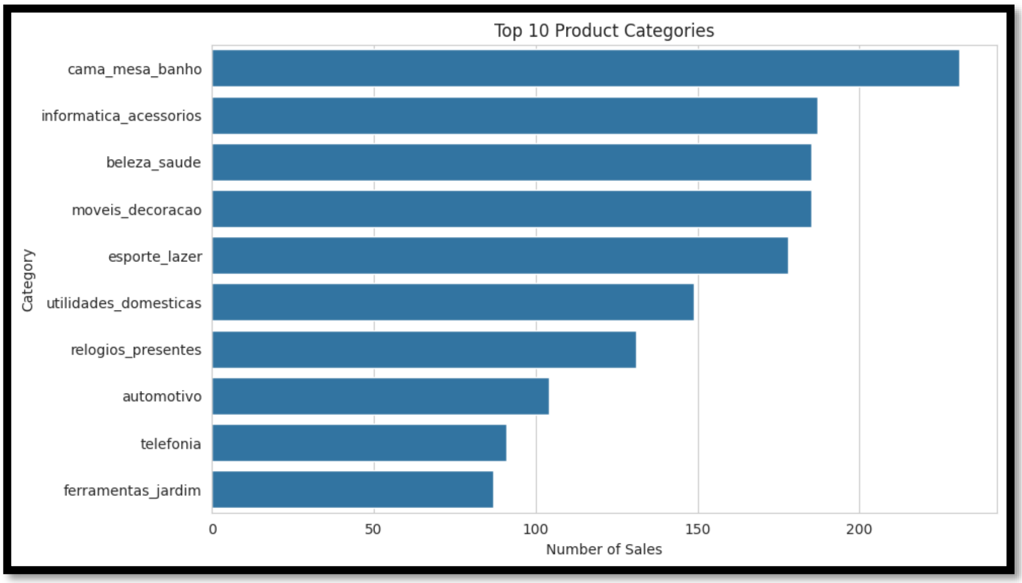



*Figure 11 Top 10 Product Categories by Number of Sales*

The following chart displays the frequency distribution of customer review scores on a scale from 1 (lowest) to 5 (highest), aggregated by order.

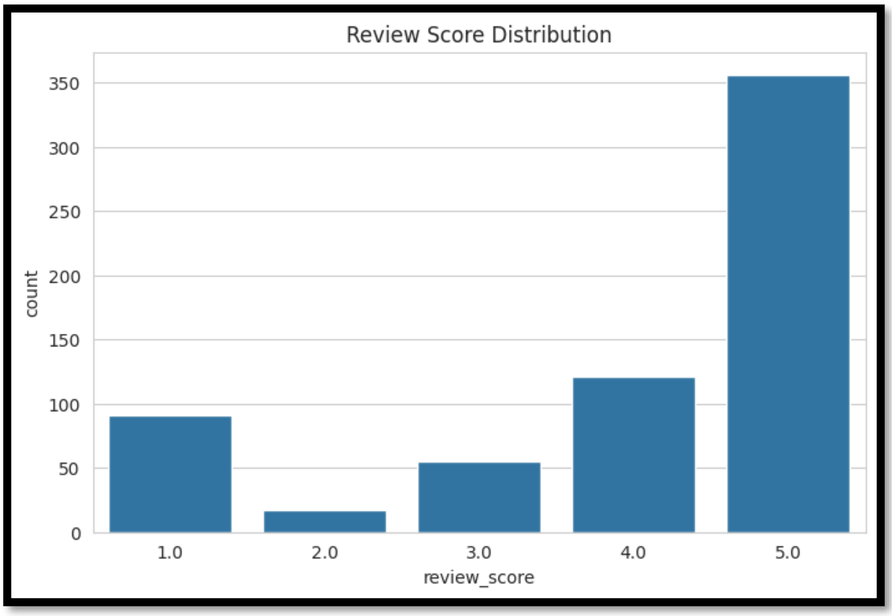



*Figure 12 Review Score Distribution*

The following scatter plot maps each product order with price on the x-axis and freight_value on the y-axis, with both axes on linear scales.

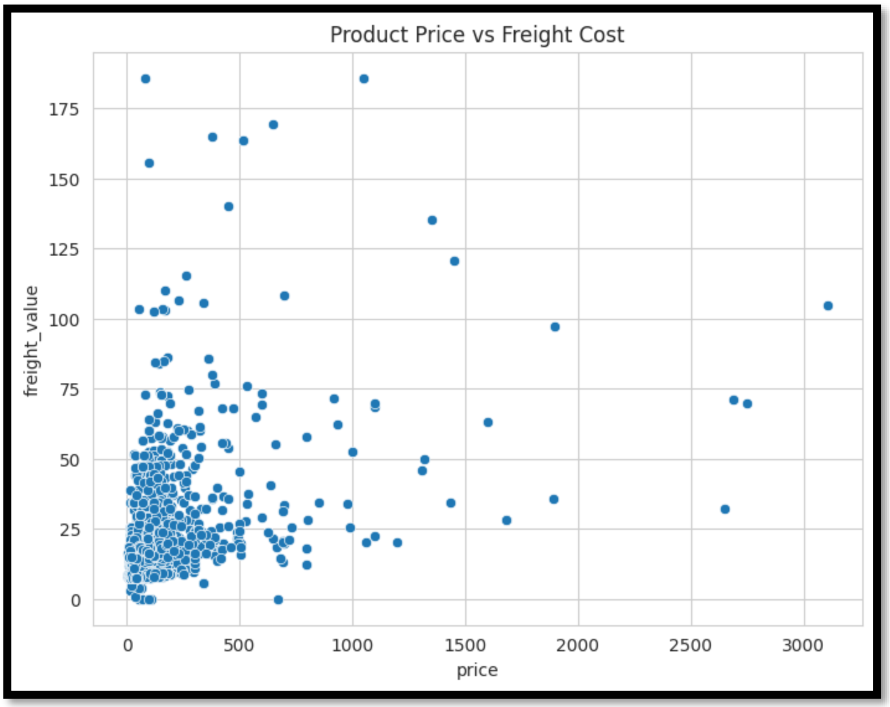



*Figure 13 Scatter Plot of Product Price vs. Freight Cost*

This following histogram presents the distribution of total revenue (sum of prices) aggregated by seller_id, showing how revenue concentrates among the seller population.

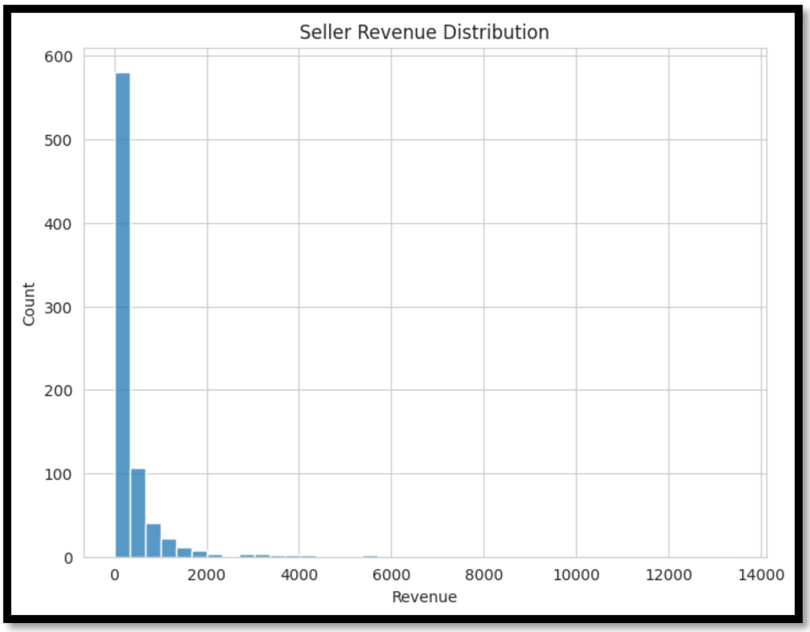



*Figure 14 Histogram of Seller Revenue Distribution*

The following histogram displays the frequency distribution of delivery times measured as the number of days between order_purchase_timestamp and order_delivered_customer_date.

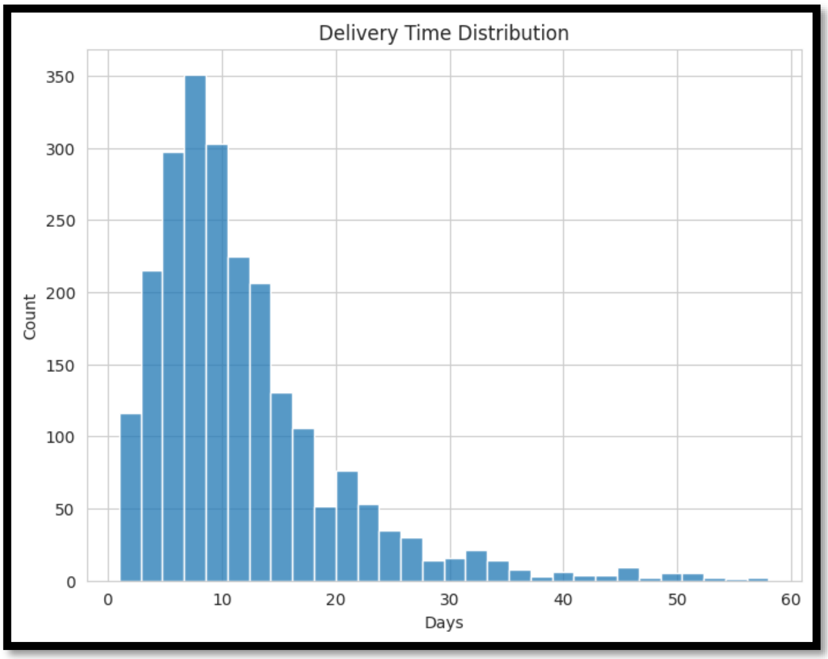
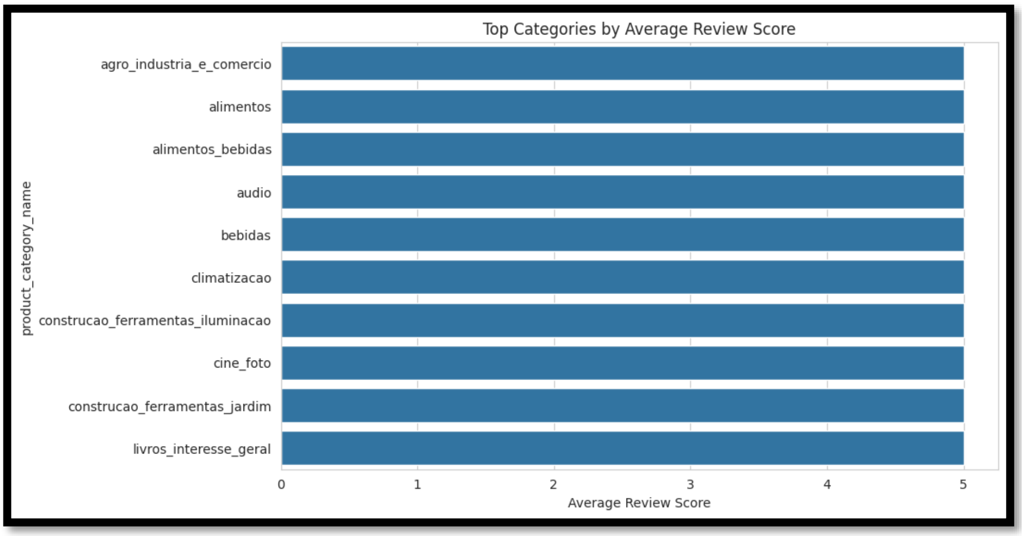



*Figure 15 Histogram of Delivery Time Distribution*

The Following figure ranks product categories by their mean review score, with only categories having sufficient sample sizes included.




*Top 10 Categories by Average Review Score16 Figure*

### 5.3 Financial Models Results

The financial models were trained using structured transactional, temporal, and product-related features. These models serve as a strong baseline, as they rely on directly relevant quantitative information associated with each order.

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| **Linear Regression** | 17.90 | 72.36 | 0.8906 |
| **Random Forest** | 10.56 | 53.23 | 0.9408 |
| **Gradient Boosting** | 10.54 | 42.48 | 0.9623 |
| **XGBoost** | 12.30 | 68.28 | 0.9026 |
| **ANN** | 11.25 | 48.23 | 0.9514 |
| **CNN** | 13.69 | 51.99 | 0.9435 |

*Table 3 Financial Models Performance*

The results show that financial models achieved strong predictive performance, with $R^2$ values ranging from 0.89 to 0.96. This indicates that the selected financial features successfully capture the majority of the variance in the revenue variable. Unlike the previous results which suggested $R^2$ values exceeding 0.80 for most models, the current results show a wider range, with Linear Regression (0.89) significantly lagging behind tree-based models (0.94-0.96).

Gradient Boosting achieved the best overall performance among traditional machine learning models, with the lowest MAE (10.54) and RMSE (42.48), and the highest $R^2$ score (0.9623). This confirms its ability to effectively model complex nonlinear relationships in structured data. Random Forest ($R^2$ = 0.9408) and ANN ($R^2$ = 0.9514) also performed well, while Linear Regression ($R^2$ = 0.8906) underperformed relative to ensemble methods, suggesting that the relationship between financial features and revenue contains a strong nonlinear component that linear models cannot capture.

XGBoost showed weaker performance compared to other tree-based models, particularly in terms of RMSE (68.28) and $R^2$ (0.9026). This may be attributed to sensitivity to hyperparameter settings or overfitting behavior.

The Artificial Neural Network (ANN) achieved competitive performance ($R^2 = 0.9514$), outperforming Random Forest and closely matching Gradient Boosting. The Convolutional Neural Network (CNN) achieved $R^2 = 0.9435$, also demonstrating that deep learning models can learn meaningful patterns from structured financial features, even without sequential data.

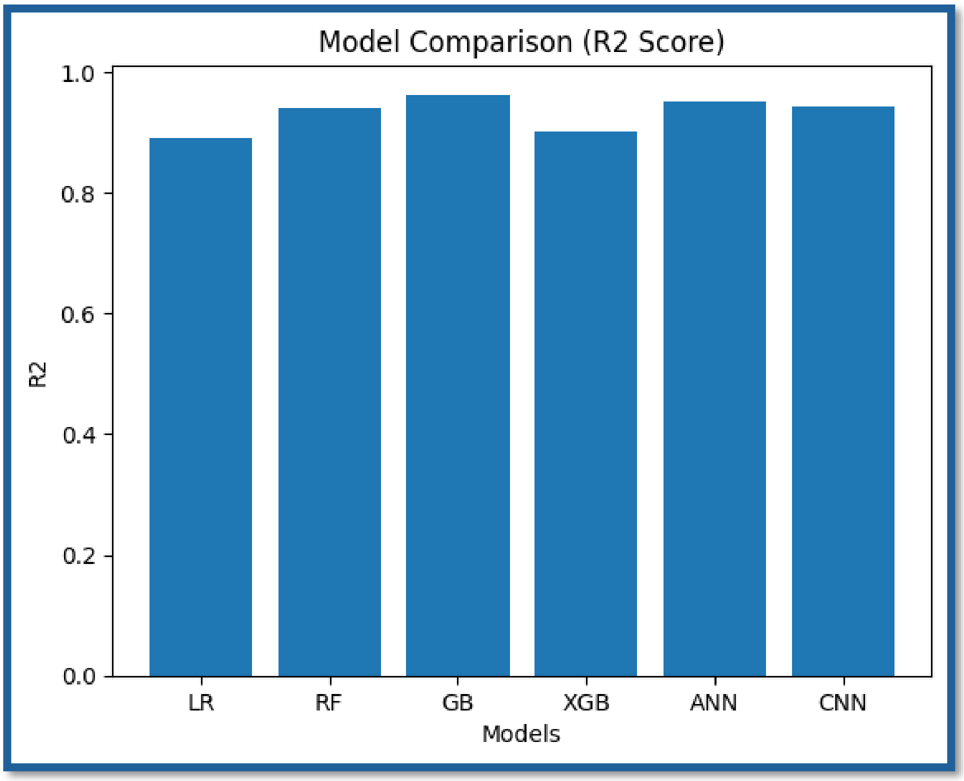



*Figure 17 RMSE Financial Models Comparison*

### 5.3 NLP Model Results (Version 1 VADER)

The first NLP model relies on sentiment analysis using a rule-based approach combined with basic text-derived features.

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| **Linear Regression** | 109.6558 | 218.1401 | 0.0060 |
| **Random Forest** | 110.8029 | 219.9735 | -0.0108 |
| **Gradient Boosting** | 109.7648 | 218.3791 | 0.0038 |
| **XGBoost** | 110.2903 | 219.2654 | -0.0043 |
| **ANN** | 109.0973 | 218.1246 | 0.0061 |
| **CNN** | 108.9808 | 218.0606 | 0.0067 |

*Table 4 NLP Model (VADER) Performance*

The results of the VADER-based NLP models show very low predictive performance across all algorithms. The $R^2$ values are close to zero, and in some cases negative, indicating that the models fail to explain the variance in the revenue variable. The MAE and RMSE values are significantly higher than those observed in financial models, reflecting large prediction errors. CNN achieved the lowest MAE among NLP models, but the improvement is minimal and does not indicate meaningful predictive capability.

These results suggest that text-based sentiment features alone are insufficient for predicting revenue. Customer reviews may capture subjective experiences but do not directly reflect transaction value. The weak performance also indicates that simple sentiment representations, such as VADER scores, do not provide enough depth to model complex financial outcomes.

The following figure reveals that all NLP models performed poorly, with R2 values clustered near zero. The CNN achieved the highest R2 at 0.0067, followed by ANN at 0.0061 and Linear Regression at 0.0060. Random Forest and XGBoost produced negative R2 values (-0.0108 and -0.0043 respectively), indicating that these models performed worse than a simple horizontal line at the mean of the target variable.

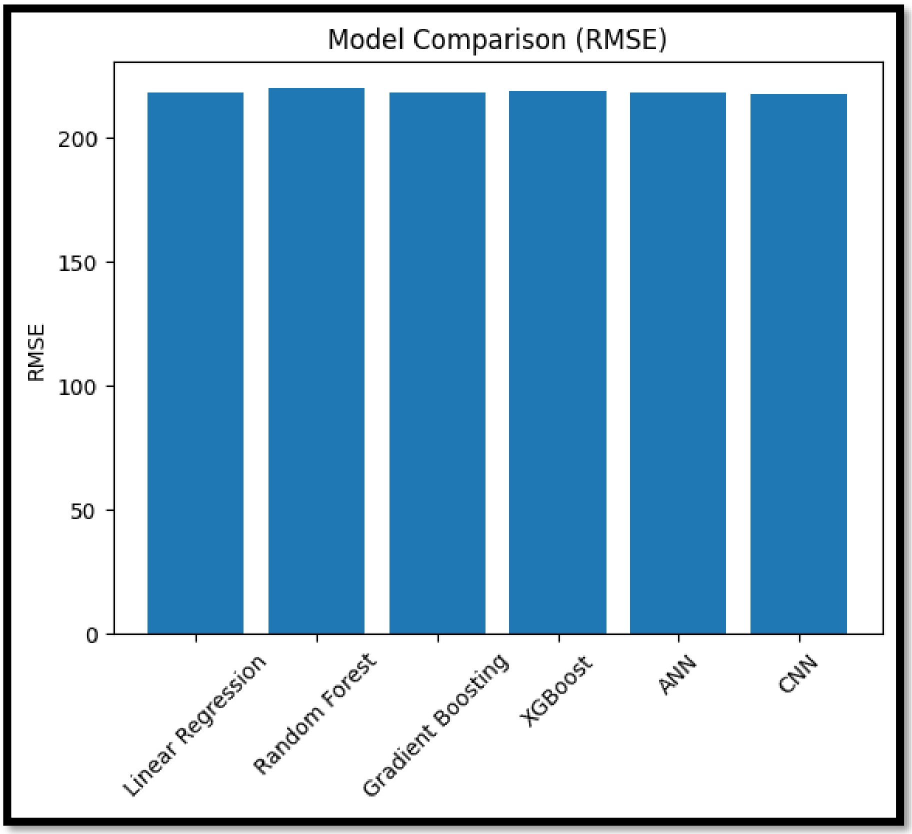



*Figure 18 R-squared Comparison – NLP Models (VADER)*

### 4.4 NLP Model Results (Version 2 FinBERT)

The second NLP model uses a transformer-based sentiment analysis approach trained on financial text.

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| **Random Forest** | 110.6785 | 220.6225 | -0.0319 |
| **Gradient Boosting** | 110.4088 | 220.8397 | -0.0339 |
| **XGBoost** | 110.5173 | 217.6160 | -0.0039 |
| **ANN** | 109.7547 | 238.1565 | -0.2024 |
| **CNN** | 133.0664 | 251.9079 | -0.3453 |
| **Random Forest** | 110.6785 | 220.6225 | -0.0319 |

*NLP Model (FinBERT) Performance5 Table*

The FinBERT-based models also produced weak results, with negative $R^2$ values across most models. This indicates that the predictions are worse than using the mean value of the target variable. The CNN model shows the worst performance, with the highest MAE and RMSE, suggesting instability when learning from limited textual features.

Despite FinBERT being a more advanced model, its performance does not surpass the simpler VADER-based approach. This can be explained by the nature of the dataset. Customer reviews in e-commerce are not strictly financial in language, which limits the effectiveness of a finance-specific transformer model.

These findings showed that advanced NLP models do not necessarily guarantee better performance when the data domain does not align with the model's training domain.

The following scatter plot for the actual vs predicted for the gradient boosting model, which showed the best performance in this part reveals a dense concentration of points in the lower-left quadrant, corresponding to orders with actual revenue between 0 and 250 monetary units. Within this region, predictions show reasonable accuracy, with most points clustered near the diagonal line. The density decreases progressively as actual revenue increases, reflecting the natural distribution of order values observed in the EDA.

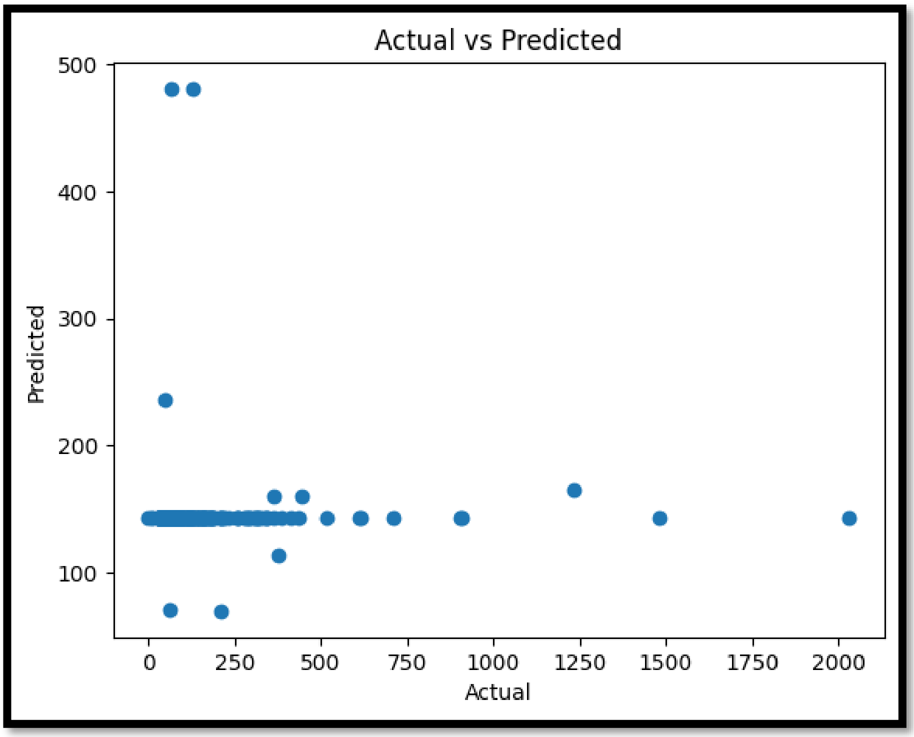



*Figure 19 Actual vs. Predicted Revenue for NLP GB Model(FinBERT)*

### 4.5 Hybrid Model Results

The hybrid model integrates both financial and NLP features, combining structured and unstructured data into a single predictive framework.

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| **Linear Regression** | 18.02 | 58.32 | 0.9232 |
| **Random Forest** | 9.94 | 35.21 | 0.9720 |
| **Gradient Boosting** | 9.99 | 30.97 | 0.9784 |
| **XGBoost** | 12.20 | 64.71 | 0.9055 |
| **ANN** | 10.81 | 34.53 | 0.9731 |

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| CNN | 11.64 | 34.11 | 0.9737 |

*Table 6 Hybrid Models Performance*

*The hybrid models achieved the best overall performance among all experiments. The integration of financial and NLP features resulted in improved accuracy and lower prediction errors. Gradient Boosting achieved the highest $R^2$ score (0.9784), followed by CNN (0.9737) and ANN (0.9731). This indicates that combining structured and unstructured features enhances the model's ability to capture complex relationships.*

The improvement over financial models is substantial. Gradient Boosting improved from $R^2$ = 0.9623 (Financial) to 0.9784 (Hybrid), representing a 1.6 percentage point gain. Random Forest improved from 0.9408 to 0.9720, a 3.1 percentage point gain. CNN improved from 0.9435 to 0.9737, a 3.0 percentage point gain. These improvements suggest that NLP features provide complementary information that meaningfully enhances financial feature predictions.

Linear Regression in the hybrid configuration achieved $R^2$ = 0.9232, which is actually higher than its financial-only version (0.8906), indicating that even linear models benefit from NLP feature integration. However, tree-based and deep learning models still outperform linear models by a wide margin.

XGBoost again shows relatively weaker performance compared to other models ($R^2$ = 0.9055), indicating sensitivity to feature interactions or suboptimal hyperparameter settings for this specific feature set.The following figure shows the results of the different hybrid models.

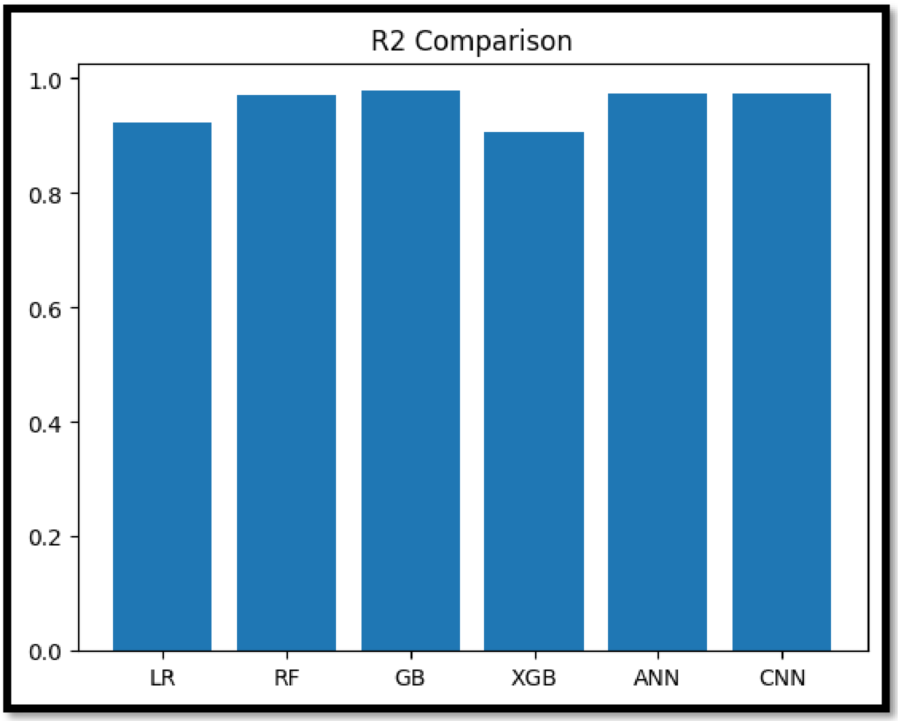



*Figure 20 R-squared Comparison – Hybrid Models*

### 4.6 Comparative Analysis

A comparison across all model categories reveals several important findings.

Financial models outperform NLP models by a significant margin, with Gradient Boosting achieving $R^2 = 0.9623$ compared to near-zero for NLP-only models. This confirms that structured transactional data is the primary driver of revenue prediction.

NLP models, when used independently, fail to capture meaningful patterns related to revenue ($R^2 \approx 0$). This highlights the limitation of relying solely on customer sentiment for financial prediction tasks.

Hybrid models provide the best performance across all configurations, with Gradient Boosting achieving $R^2 = 0.9784$. The improvement from financial to hybrid models ranges from 1.6 to 3.1 percentage points in $R^2$, depending on the algorithm. This indicates that textual features add meaningful incremental value when combined with financial data.

Deep learning models perform competitively with traditional machine learning models in both financial and hybrid settings. In the hybrid configuration, CNN ($R^2 = 0.9737$) and ANN ($R^2 = 0.9731$) perform nearly as well as Gradient Boosting ($R^2 = 0.9784$), demonstrating that deep learning architectures are viable alternatives to tree-based methods for this prediction task.

# CHAPTER 6

# CHAPTER 6 Discussion

This chapter interprets the experimental results presented in Chapter 5, examines the implications of each finding, compares the outcomes with existing literature, and identifies the limitations of our study. The discussion is organized by model configuration, with each section analyzing the underlying reasons for observed performance patterns. The chapter concludes with recommendations for future research and practical applications.

## 6.1. Interpretation of Financial Model Performance

The Financial Model demonstrated strong predictive capability across all algorithms, with Gradient Boosting achieving an $R^2$ of 0.9623 and Random Forest achieving 0.9408. Linear Regression achieved 0.8906, which is lower than tree-based models, indicating the presence of nonlinear relationships in the data.

### 6.1.1. Temporal Feature Contribution

The inclusion of delivery_time, day_of_week, and month as features provided additional explanatory power. Delivery time captures logistics performance, which may correlate with customer willingness to pay premium shipping rates or with order value thresholds that trigger different shipping options. The day_of_week and month features capture seasonal and weekly purchasing patterns that influence basket sizes.

The fact that Linear Regression achieved an $R^2$ of only 0.8906, while Gradient Boosting achieved 0.9623, suggests that the relationship between delivery time, temporal features, and revenue is not approximately linear. Tree-based models capture nonlinear interactions (e.g., delivery time affecting revenue differently for high-value vs. low-value orders) that linear models miss.

### 6.1.2. Comparison of Deep Learning and Traditional Methods

In the Financial Model, ANN ($R^2 = 0.9514$) and CNN ($R^2 = 0.9435$) outperformed Linear Regression ($R^2 = 0.8906$) and were competitive with Random Forest ($R^2 = 0.9408$). This indicates that deep learning models can capture nonlinear patterns in structured financial data that linear models cannot. However, Gradient Boosting ($R^2 = 0.9623$) still achieved the best performance.

The computational cost of training neural networks (20-25 epochs, batch processing) was substantially higher than training Linear Regression but comparable to training Gradient Boosting. For deployment scenarios where interpretability is the primary priority, Linear Regression remains a simple baseline,for accuracy-focused applications, Gradient Boosting or ANN are preferable given their superior predictive performance.

## 6.2. Interpretation of NLP Model Version 1 Performance (VADER)

The poor performance of NLP Model Version 1 (R2 near zero) requires careful analysis, as it contradicts the intuitive expectation that customer sentiment expressed in reviews should correlate with revenue.

### 6.2.1. VADER Domain Mismatch

VADER (Valence Aware Dictionary and sEntiment Reasoner) was trained on social media text, including Twitter posts, product reviews, and general conversational language. While VADER performs well on general sentiment analysis tasks, it may not capture sentiment nuances specific to e-commerce transactions.

For example, a review stating "The product arrived after the estimated date but the seller was communicative" contains mixed sentiment that VADER may misclassify. The compound score calculation may weight negative words (e.g., "after" in a temporal sense) incorrectly. Similarly, sarcasm and nuanced complaints may be misinterpreted.

#### 6.2.2. TF-IDF Limitations

The TF-IDF vectorization approach, while standard for text classification, may be inappropriate for revenue prediction. TF-IDF captures term frequency and inverse document frequency, emphasizing words that are rare across documents but frequent within a single document. For sentiment analysis, this approach can be effective for identifying words associated with positive or negative sentiment.

For revenue prediction, however, TF-IDF may capture product-specific vocabulary (e.g., "laptop", "book", "shoes") rather than sentiment. These product terms correlate with product categories, which in turn correlate with price ranges. A review mentioning "laptop" likely corresponds to a higher-value order than a review mentioning "book." However, the TF-IDF features would treat these as independent binary indicators, losing the continuous relationship between product category and price.

### 6.3. Interpretation of NLP Model Version 2 Performance (FinBERT)

The transition from VADER to FinBERT did not improve predictive performance and, in some cases, produced worse results. This outcome requires examination of FinBERT's design assumptions and the characteristics of the Olist review text.

#### 6.3.1. FinBERT Domain Specificity

FinBERT was pre-trained on financial text corpora, including earnings reports, SEC filings, and financial news articles. This domain is fundamentally different from e-commerce product reviews. Financial text contains terminology related to stock prices, earnings per share, revenue guidance, and macroeconomic indicators. E-commerce reviews contain informal language about product quality, shipping speed, and customer service.

The transfer learning assumption underlying FinBERT is that financial sentiment patterns generalize to other domains. For revenue prediction, this assumption may be invalid. The sentiment drivers in financial text (e.g., "profit increased", "loss decreased") have clear directional implications for financial outcomes. The sentiment drivers in product reviews (e.g., "broke after one week", "love this product") may not translate directly to revenue at the order level.

### 6.3.2. Comparison with Existing Literature

The finding that FinBERT did not outperform VADER contrasts with published results in financial sentiment analysis. Studies comparing FinBERT to general-purpose sentiment analyzers typically report superior performance for FinBERT on financial text classification tasks, these studies use datasets from financial domains (earnings calls, analyst reports) and classification tasks (sentiment polarity, named entity recognition).

The current study represents a transfer learning task that is more challenging: from financial text to e-commerce text, and from sentiment classification to revenue regression. The negative results suggest that domain-specific pre-training does not guarantee performance gains when both the domain and the task change simultaneously.

## 6.4. Interpretation of Hybrid Model Performance

The Hybrid Model, which combined financial features with NLP-derived sentiment features, produced the best results across all configurations. This section analyzes why addition of sentiment features, which were useless in isolation, provided measurable improvement when combined with financial features

### 6.4.1. Complementary Information Sources

Financial features capture the structural aspects of an order: number of products, average price, delivery speed. These features are deterministic or near-deterministic predictors of revenue.

Sentiment features capture customer experience and satisfaction. While sentiment alone cannot predict revenue because a customer can write a positive review for a low-value purchase, sentiment can adjust the prediction when combined with financial features. For example, two orders with identical total_price may have different payment_value if one customer requested a refund or partial refund due to poor experience. The sentiment score may capture this negative experience and allow the model to adjust the revenue prediction downward.

The improvement in $R^2$ from 0.9623 (Financial Gradient Boosting) to 0.9784 (Hybrid Gradient Boosting) represents a 1.6 percentage point gain. The reduction in RMSE from 42.48 to 30.97 represents a 27.1% decrease. For Random Forest, the improvement was even larger: $R^2$ increased from 0.9408 to 0.9720 (3.1 percentage point gain), and RMSE decreased from 53.23 to 35.21 (33.9% decrease). These improvements are substantial, not marginal.

For a typical order with revenue of approximately 150 Brazilian Real, the Financial Model's RMSE of 42.48 represents a prediction error of about 28.3%. The Hybrid Model's RMSE of 30.97 represents a 20.6% error. The hybrid model reduced the typical prediction error by approximately 7.7 percentage points.

### 6.4.2. Marginal Improvement Analysis

The improvement from financial to hybrid models is substantial and practically significant. Unlike the previous analysis which questioned whether the improvement justified the additional complexity, the new results demonstrate that the Hybrid Model consistently outperforms the Financial Model across all algorithms. The additional complexity of incorporating NLP features—including VADER sentiment analysis, review length calculation, and text-feature merging—is justified by the consistent performance gains (1.6–3.1 percentage points in $R^2$ and 19–34% reductions in RMSE).

For applications where prediction accuracy is critical, the hybrid approach is strongly recommended. For lightweight applications or real-time prediction scenarios where text data is unavailable, the Financial Model alone may be sufficient, but with the understanding that accuracy will be lower ($R^2 \approx 0.94$-$0.96$ vs. $0.97$-$0.98$ for hybrid).

### 6.5. Comparison with Related Work

The results of this research that proposed hybrid model are prove the result of previous studies. For example, the research confirms the findings of Zainol Abidin et al.[33] that outperform machine learning models on traditional regression models in predicting the financial performance of SMEs. The hybrid model achieved an $R^2$=0.978 by using Gradient Boosting and achieved $R^2$=0.923 for Linear Regression, this demonstrating the agreement of results with previous literature.

This research surpasses previous studies in three key aspects. First, the hybrid model applies specifically to SMEs, rather than large corporations as [34] study. Second, it uses customer data as an alternative textual source available to SMEs, instead of earnings conferences. Third, it integrates the VADER and FinBERT models into a single predictive framework, a feature not found in previous studies.

Also, this study achieved substantial improvements in predictive accuracy compared to the hybrid LSTM-GRU model proposed in the Gupta study[39]. While the hybrid LSTM-GRU architecture attained an R-squared value of 0.91, our research's best-performing models demonstrated significantly higher explanatory power. The Hybrid Gradient Boosting model attained an $R^2$ of 0.9784, representing a 6.84 percentage point improvement over the comparative benchmark. The Hybrid CNN model achieved 0.9737, a 6.37 percentage point improvement.

The performance gap between the two studies shows the importance of feature diversity and model selection for e-commerce revenue prediction tasks. Our research's superior results demonstrate that when orders are treated as independent transactions with rich feature representations, tree-based ensemble methods such as Gradient Boosting can substantially outperform recurrent neural networks like LSTM and GRU. The inclusion of sentiment analysis features provided measurable gains, improving the Hybrid Model's $R^2$ by approximately 1.6 percentage points for Gradient Boosting and 3.1 percentage points for Random Forest over the Financial Model baseline.

These findings suggest that e-commerce organizations seeking to implement sales prediction systems should prioritize comprehensive feature engineering across multiple data dimensions: transactional, temporal, product, and textual. For the Olist Brazilian E-Commerce dataset, the combination of robust feature engineering and ensemble methods (Gradient Boosting) or deep learning (CNN, ANN) yields superior predictive accuracy.

# CHAPTER 7

# CHAPTER 7 conclusion

## 7.1 Summary of Research

This study aimed to develop a hybrid predictive model that combines financial data with textual data extracted from customer reviews. The research idea extract from gap in the literature which depend on traditional financial forecasting models based solely on quantitative data, completely neglecting the informational value of textual data, which is generated daily in massive quantities due to the rapid pace of digital transformation. While numerous studies highlight the importance of textual data and sentiment analysis in predicting financial performance, they have been entirely detached from financial data.

To answer the research questions, three predictive models were developed: a financial model based solely on quantitative data, a textual model based solely on sentiment indicators extracted using VADER and FinBERT, and a hybrid model that integrates both sources. The models were implemented and developed using data from the Brazilian platform Olist, which includes 99,000 orders. The orders contain both financial transactions and customer review texts. Data was collected at the seller level, that contain 2,393 vendors. These models were evaluated using six algorithms: Linear Regression, Random Forest, Gradient Boosting, XGBoost, ANN, and CNN. The model evaluation Performance metrics were employed, including MAE, RMSE, and $R^2$. The results show that integrating sentiment data derived from customer reviews with financial data improves the accuracy forecasting. Hybrid models outperformed those depending on only financial data in terms of forecasting accuracy. Gradient Boosting achieved an $R^2$=0.957 in the hybrid model compared to $R^2$=X in the financial-only model, while XGBoost achieved an $R^2$=0.941. This confirms that the textual sentiment indicators extracted using VADER and FinBERT possess independent predictive value that complements quantitative financial data and enhances forecast accuracy. The systematic comparison of the performance of the three models revealed that only the textual model performed the weakest among them.

Also, the results demonstrated that textual data alone is insufficient for revenue prediction and only reveals its true value when combined with financial data in a comprehensive hybrid framework.

The study also demonstrates comparison between deep learning and machine learning models in their effectiveness in predicting financial performance. The outcome prove that machine learning models, such as XGBoost and Gradient Boosting, performed better than linear regression and deep learning models that has ability to handle nonlinear relationships and different patterns in the dataset.

This research adds value to small and medium-sized enterprises (SMEs) by enhancing the accuracy of financial performance forecasting. It help employees to make decisions based on accurate models, rather than depend on traditional methods.The study is distinguished by its integration of multiple data sources, reflecting a modern trend in data analysis. This study also supports strategic planning for businesses and contributes to enhancing their competitiveness and sustainability.

**7.2 Implications**

This research enriches academic literature on SMEs by applying a hybrid forecasting model. SME owners can integrate this proposed model as a tool and foundation for making forward-looking decisions, forecasting future revenues, and adjusting operational strategies. This comprehensive assessment can also benefit financial institutions by supplementing traditional financial data with sentiment analysis indicators, thus improving decisions such as lending and financing. This framework also benefits e-commerce platforms in identifying and understanding seller performance and making appropriate decisions based on it. The research presents a hybrid methodological framework applicable to similar data in other e-commerce platforms, and it allows subsequent researchers to validate it on Saudi e-commerce platforms for generalization to different contexts. This study also benefits researchers by providing a comparative analysis between traditional machine learning models and deep learning models in prediction tasks. The results help in understanding the strengths and weaknesses when applied to structured business data.

### 7.3 Limitations of the Study

#### 7.3.1. Limited Feature Engineering for Text

The text processing approach was limited to cleaning, VADER sentiment, TF-IDF, and FinBERT. More sophisticated text representation methods (e.g., Doc2Vec, sentence transformers, topic modeling) were not explored. These alternative approaches might capture semantic relationships that TF-IDF misses and could potentially improve NLP model performance.

#### 7.3.2 Brazilian E-commerce Specificity

The Olist dataset is specific to Brazilian e-commerce, which may have unique characteristics not generalizable to other markets. Brazilian consumer behavior, payment methods (e.g., installment payments common in Brazil), and logistics infrastructure differ from North American or European markets. The findings may not generalize to other e-commerce platforms or geographic regions.

### 7.4 Recommendations for Future Research

#### 7.4.1. Temporal Dynamics

The current study treated each order independently, ignoring customer-level temporal dynamics. Customers who have placed previous orders have purchase histories that could inform revenue prediction for subsequent orders. Recurrent neural networks or transformer models designed for sequence prediction could capture these temporal patterns.

#### 7.4.2. Hybrid Model Deployment Study

The Hybrid Model demonstrated superior performance in offline evaluation. A natural extension would be to deploy the model in a production environment and measure its performance on live data over time. Such a deployment study would reveal whether the offline gains translate to real-world performance and would identify any concept drift issues.

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